Data Visualization Principles and Practice: Tufte, Wilke, and Others

ACE 592 SAE

Back to Data Visualization

We’ve gone over data visualization in the course quite a bit already.

Now that we have done a bit of exploring and are more comfortable with Python, we can dive it to some general principles of making graphs.

What are the goals of data visualization?

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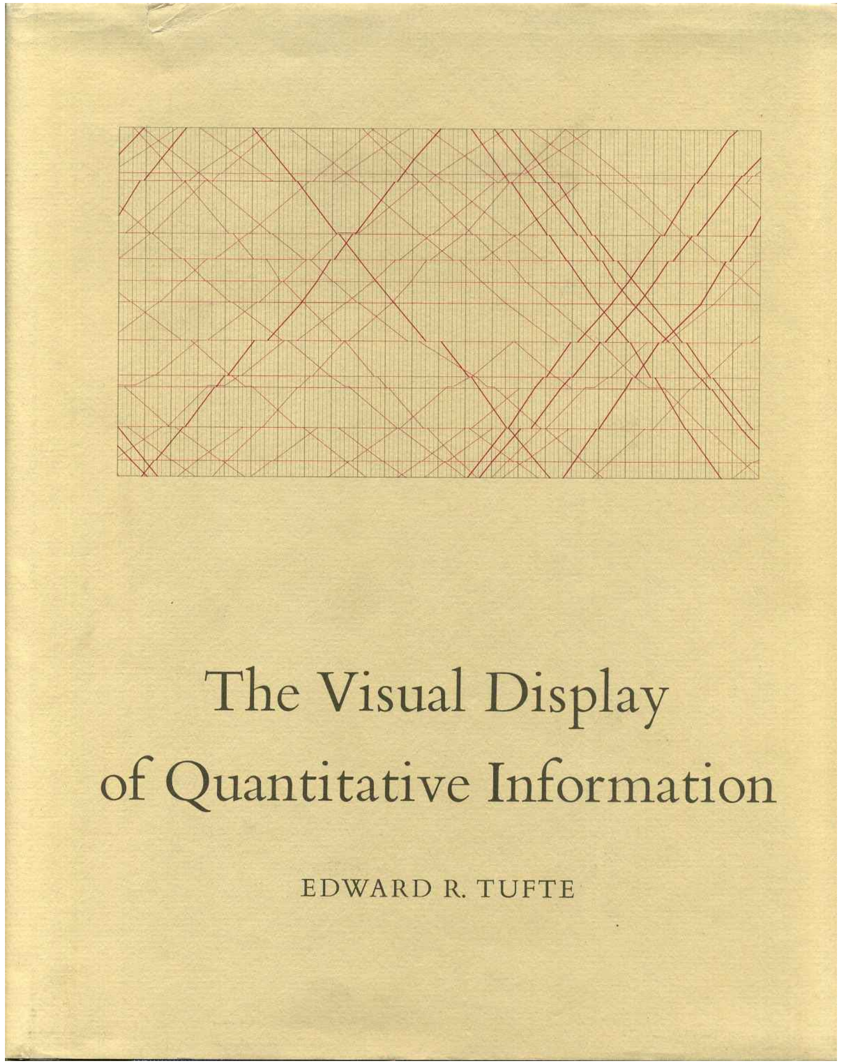
1. Tell a story!
    a. Requires you to think carefully about what sort of story you are telling.
    b. What do you want to tell your audience in one sentence?

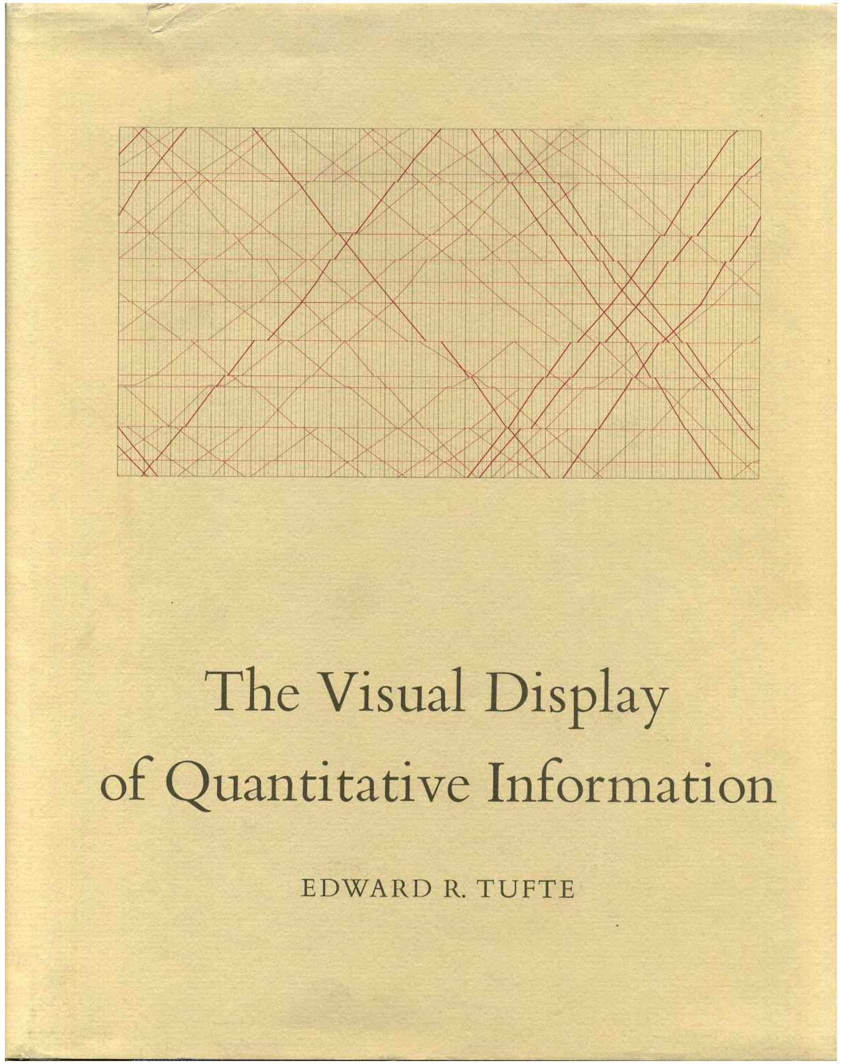
What are the goals of data visualization?

1. Tell a story!
    a. Requires you to think carefully about what sort of story you are telling.
    b. What do you want to tell your audience in one sentence?

2. Succinctly summarize data.
    a. Tables sometimes do not intuitively communicate results.
    b. Our brains are geared to sometimes take in information visually.



Data Visualization Principles: Lessons from Tufte



Edward Tufte’s book is considered a seminal text of data visualization principles.

While he has some strong (and controversial) opinions, his ***5 principles*** are useful to consider:

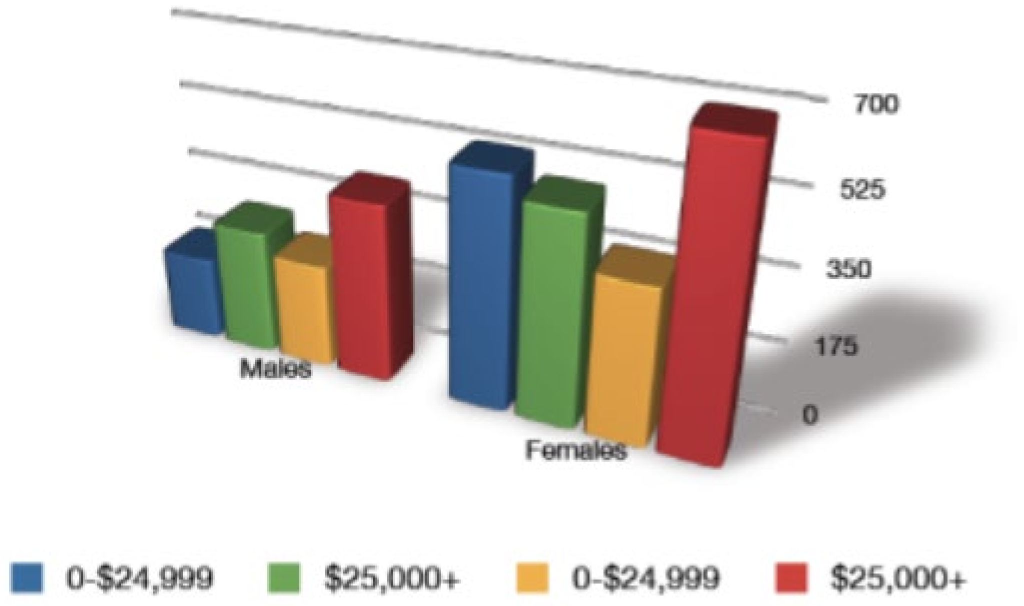
1. Above all else, show the data
2. Maximize the data-ink ratio
3. Erase non-data ink
4. Erase redundant data-ink
5. Revise and edit

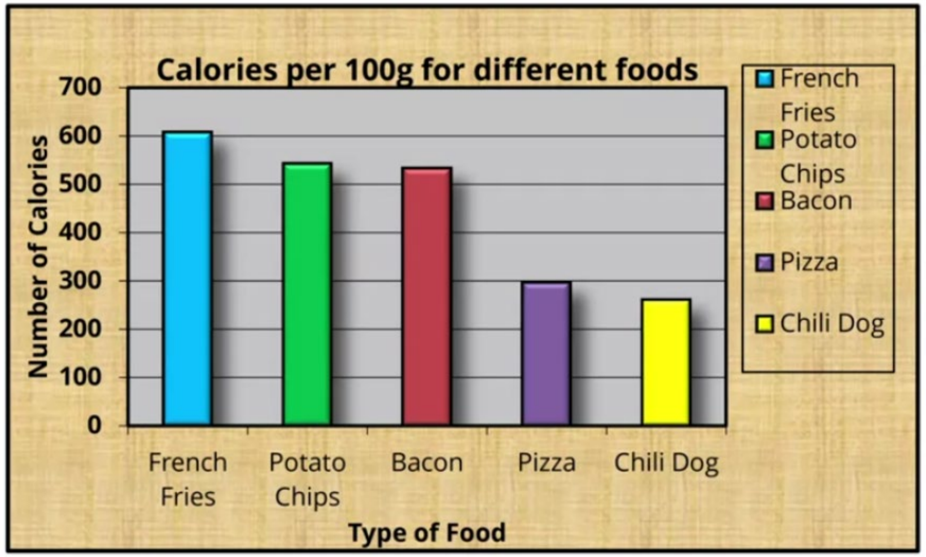
Data-ink ratio

What part of your figure is actually showing data?

Tufte specifies “data-ink” as the ink in your figure actually showing data.

Data ink ratio = data-ink/total ink





Data-ink ratio

Which of the parts of this graph below are actually necessary?

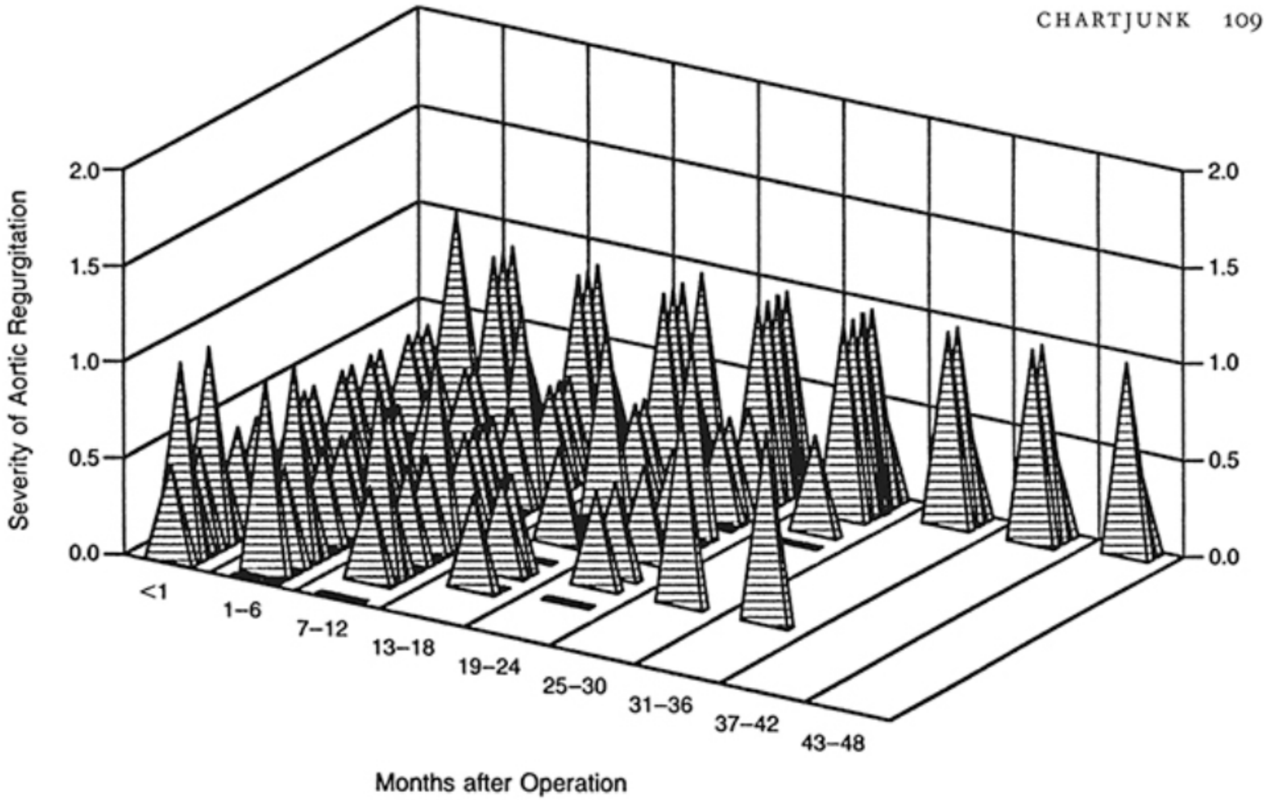


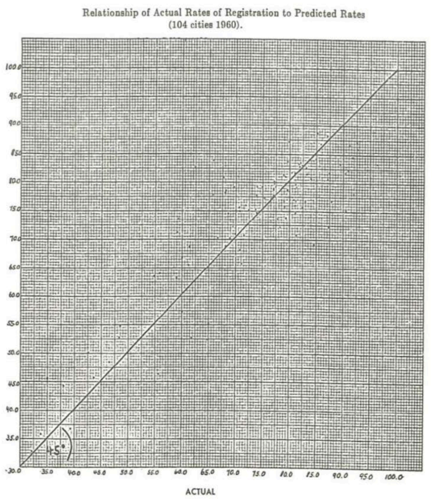
Figure 2. Serial Echocardiographic Assessments of the Severity of Regurgitation in the Pulmonary Autograft in 31 Patients. The numerical grades were assigned according to the severity of regurgitation, as follows: 0, none; 0.5, trivial; 1.0 to 1.5, mild; 2.0, moderate; and 3.0, severe.

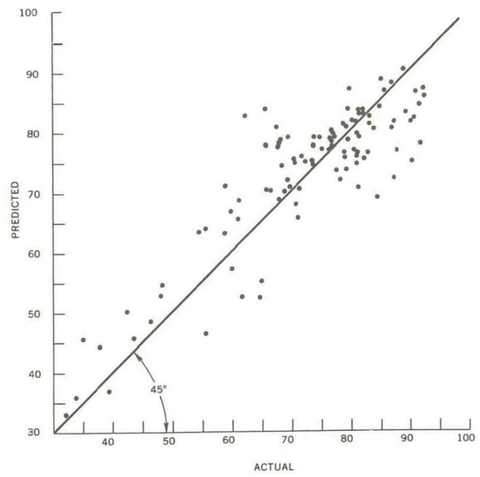
Erasing “chart junk”

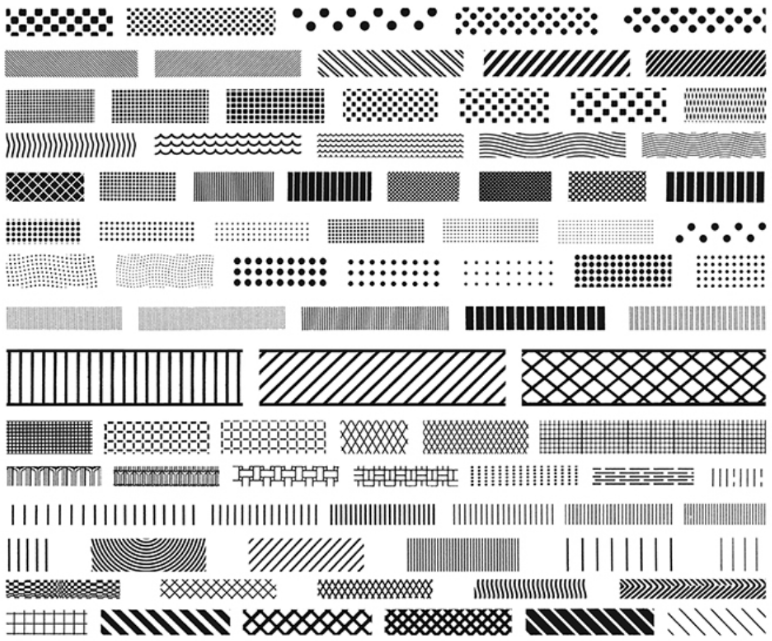
Tufte advocates for removing certain things that people would normally include:

- Grid lines
- Hatching
- Needless ornamentation

Sometimes it gets a bit radical…

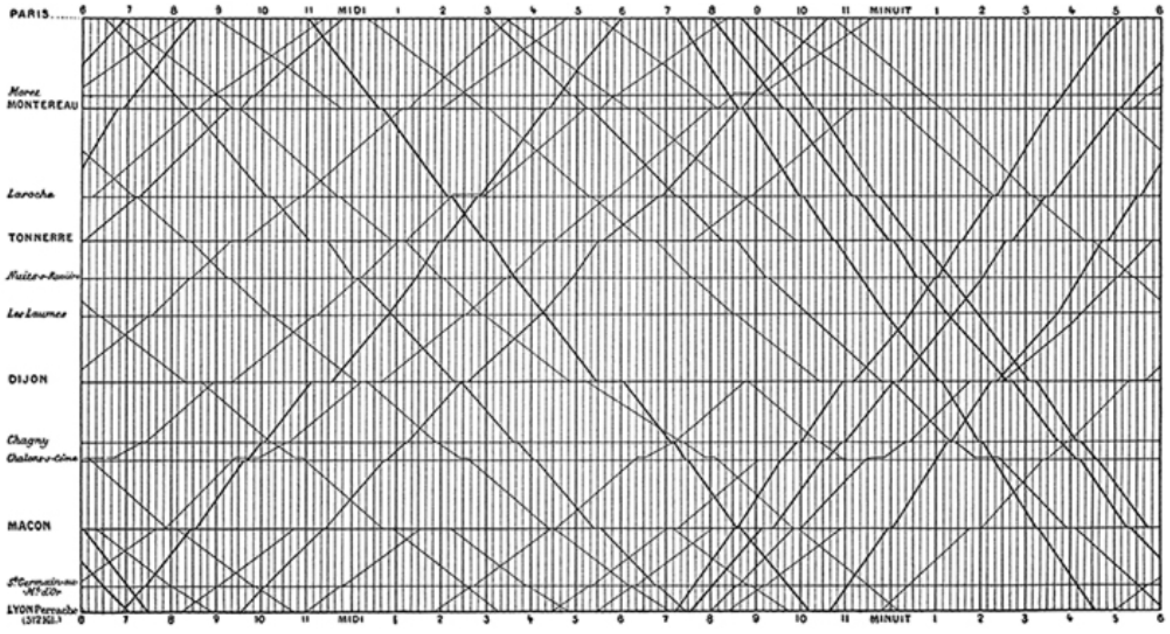




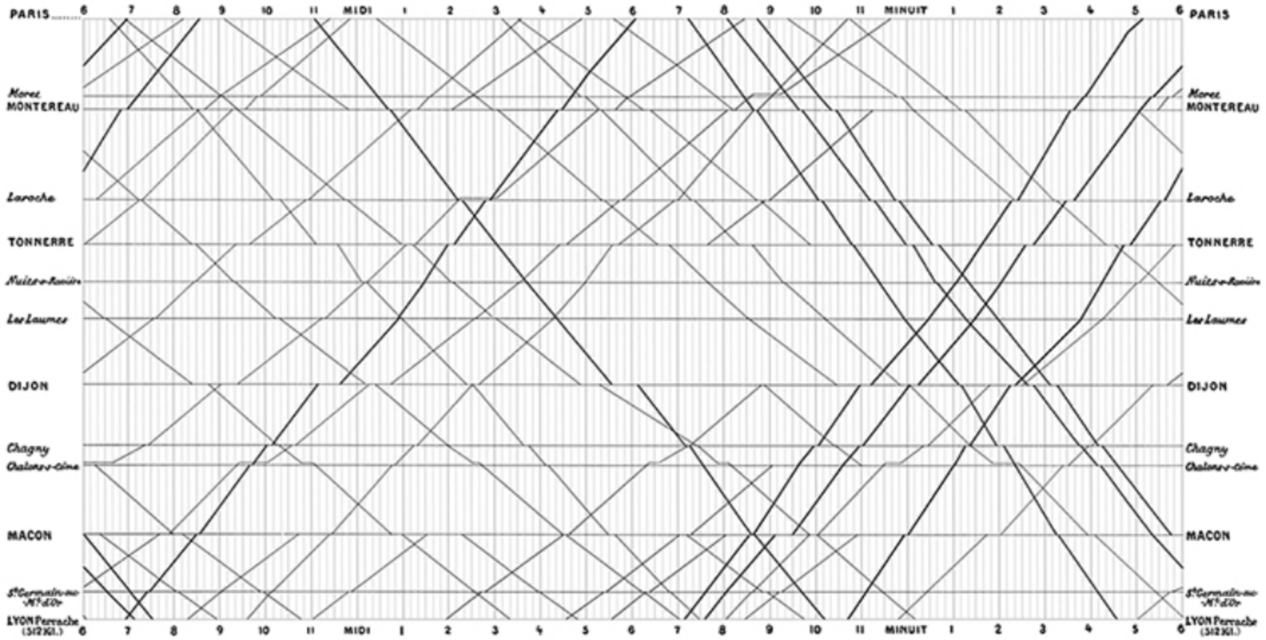


Erasing "chart junk"

The grid in the classic Marey train schedule is very active:



A better treatment, however, is a *gray grid*:

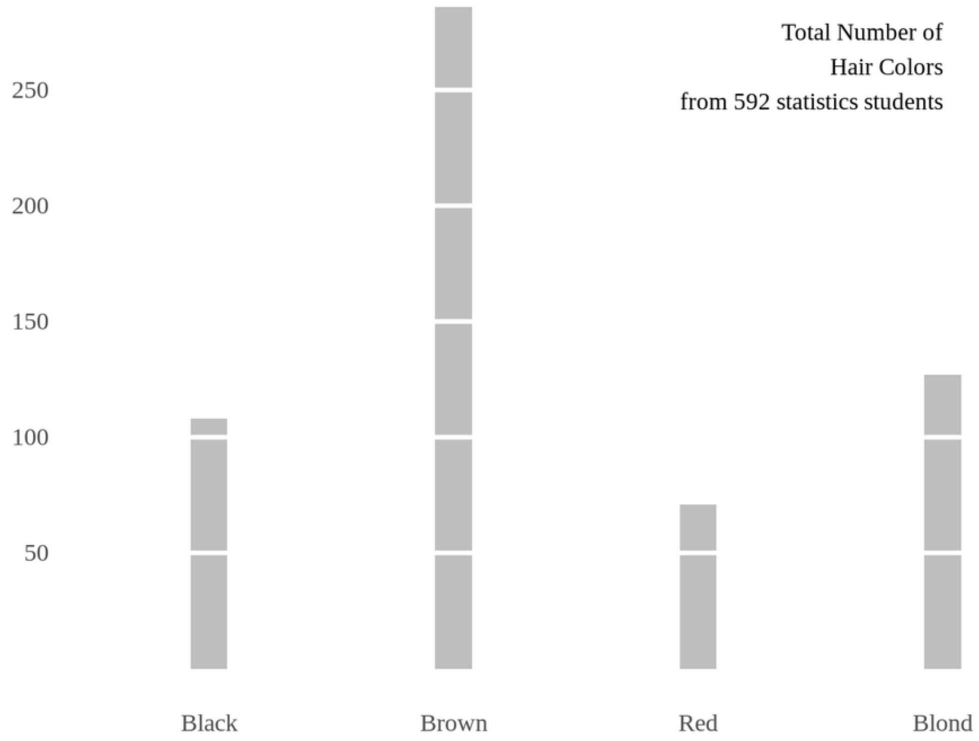


French train schedule looks better without strong gridlines!

Erasing redundant ink (within reason)

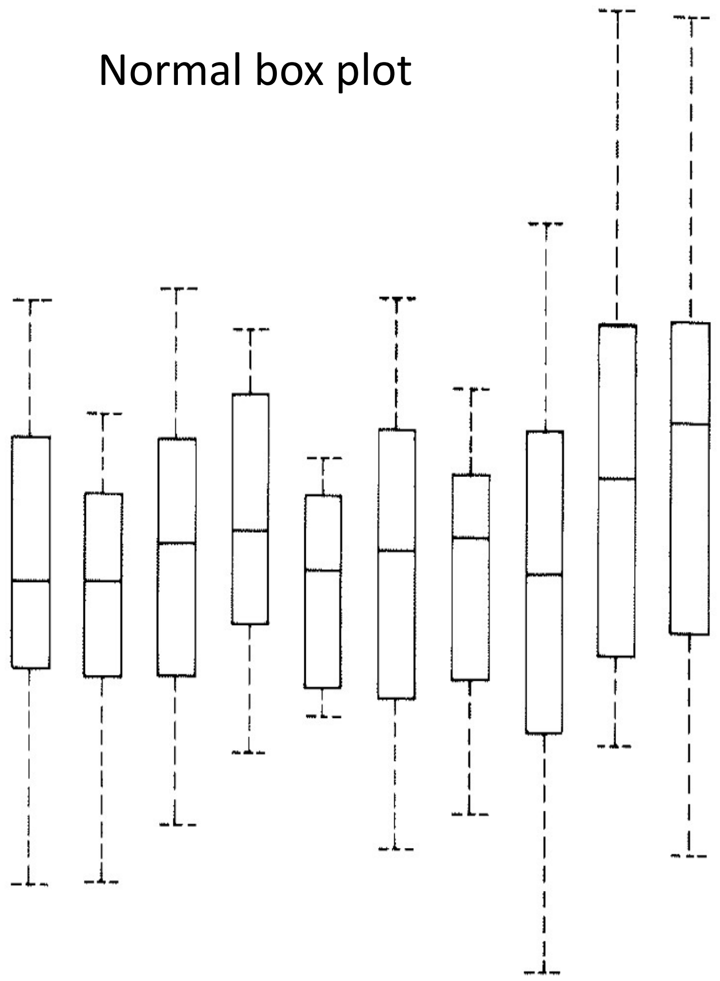
Tufte advocates going further than most people would be comfortable with, erasing things like:

- Parts of the axis not in the data
- Axes that are not being used
- Anything not strictly necessary to show the data.

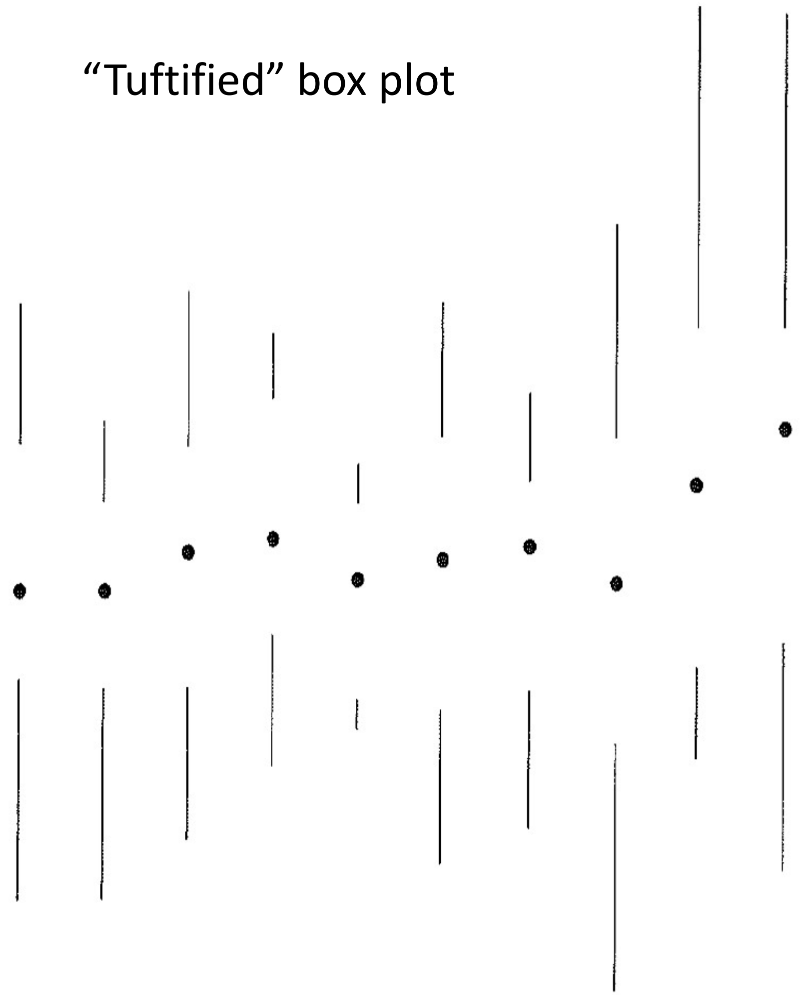


Erasing redundant ink (within reason)

Normal box plot

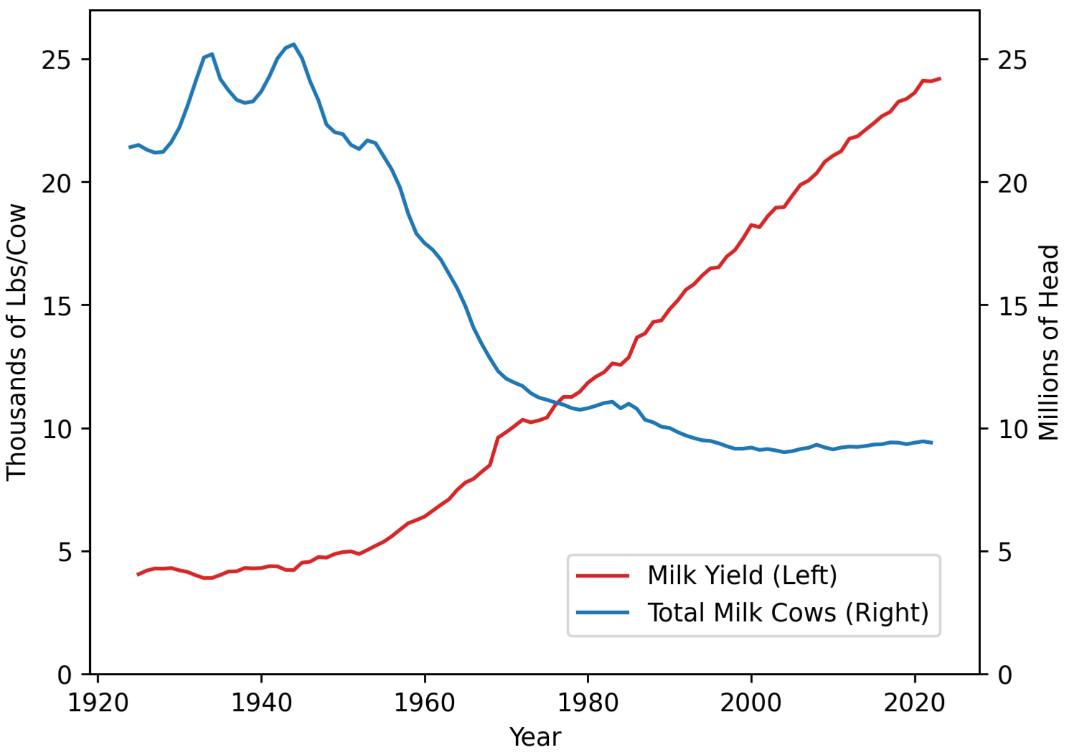


“Tuftified” box plot

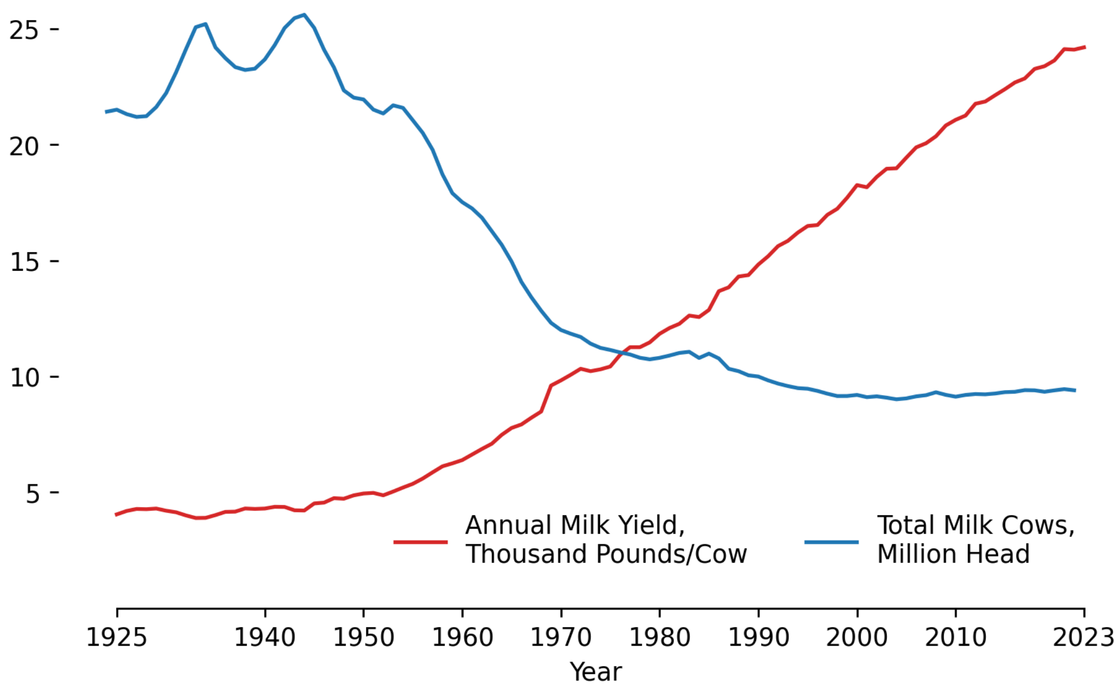


Erasing redundant ink (within reason)

My first attempt



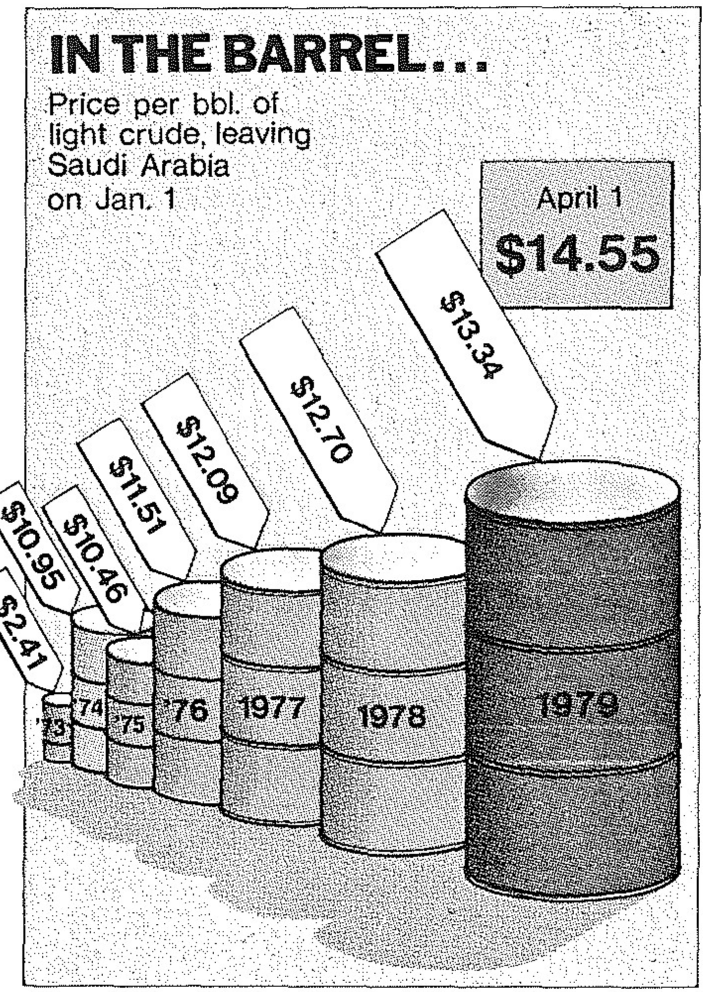
Applying Tufte edits



Proportional ink principle

“The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the numerical quantities represented”

If twice the ink, should be twice the value!

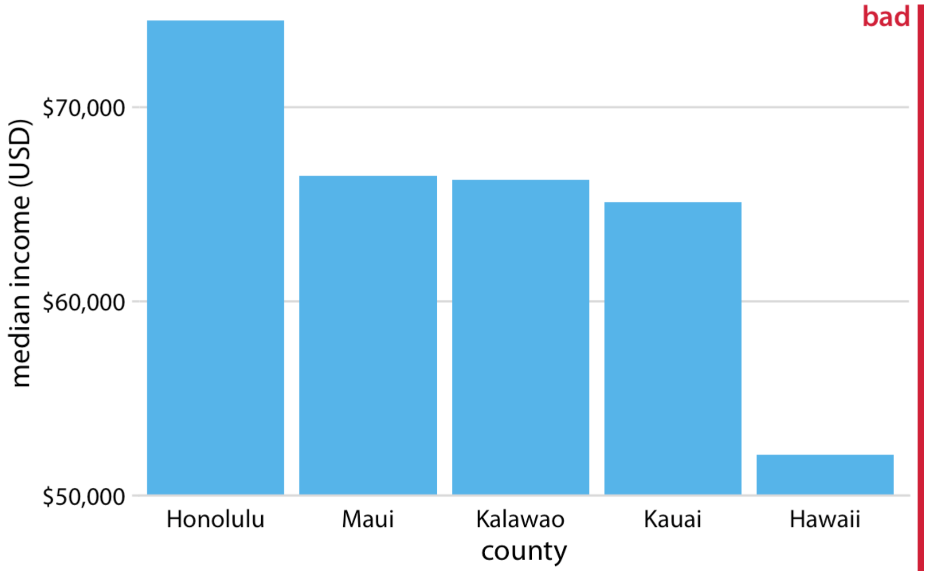


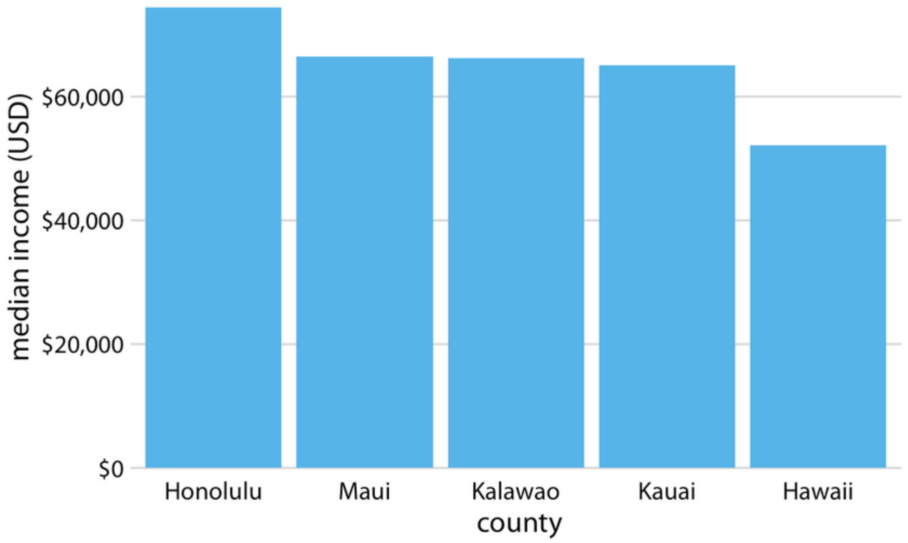
Proportional ink principle

Common ways to violate this principle:

1. Truncating the y-axis

Example

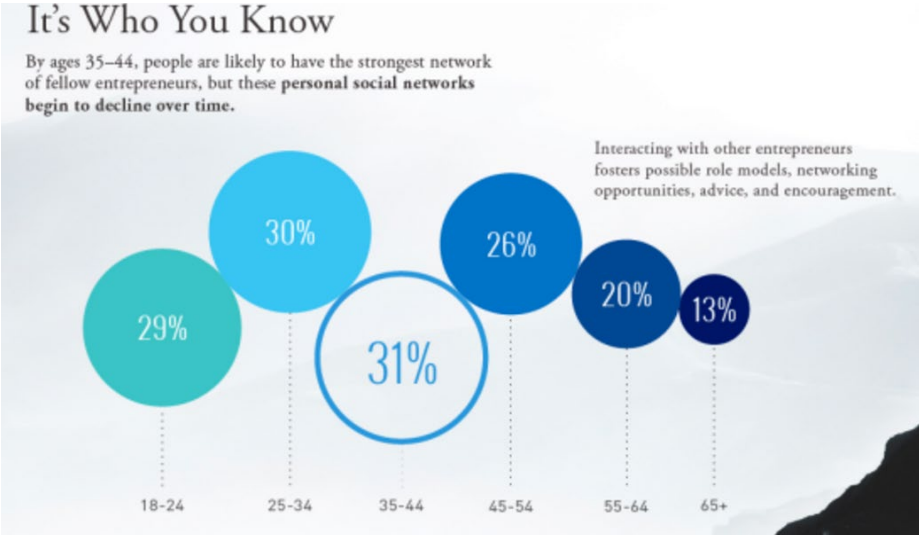


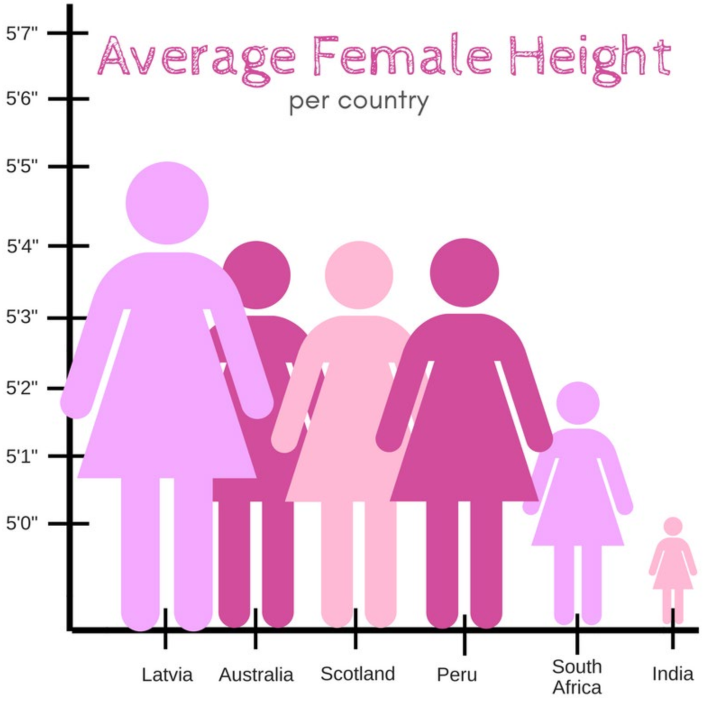


Proportional ink principle

Common ways to violate this principle:

1. Truncating the y-axis
2. Univariate data portrayed with multiple dimensions



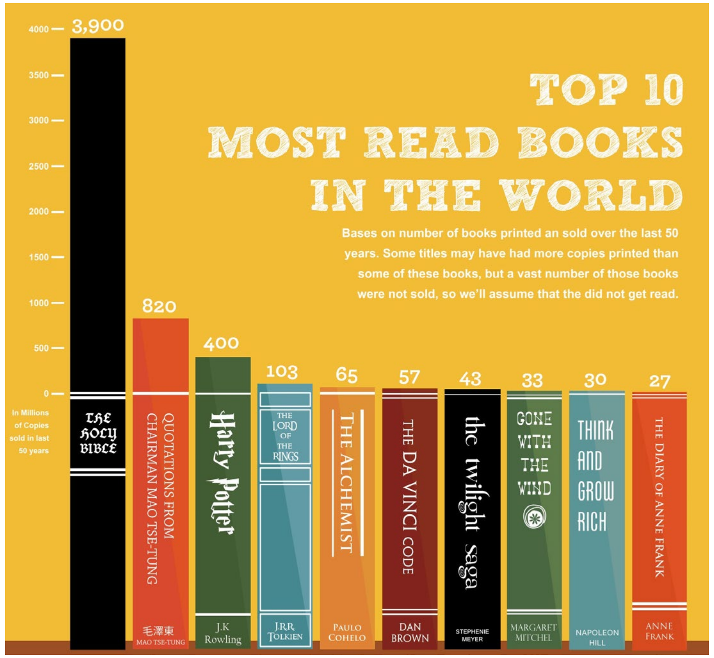


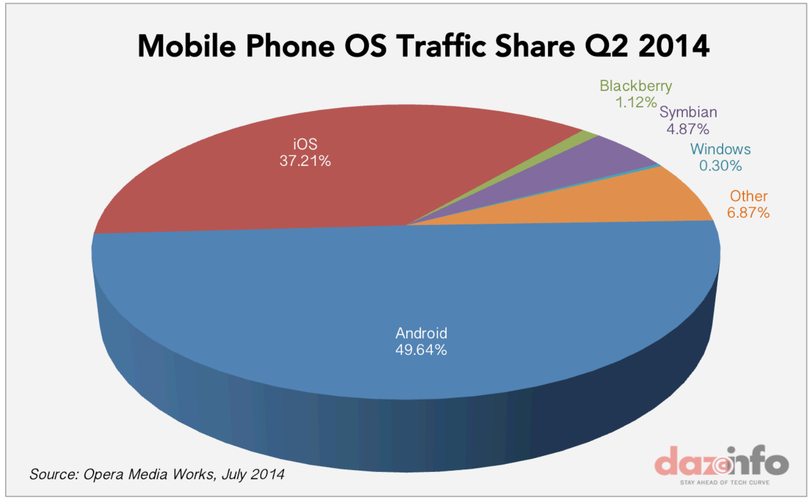
Proportional ink principle

Common ways to violate this principle:

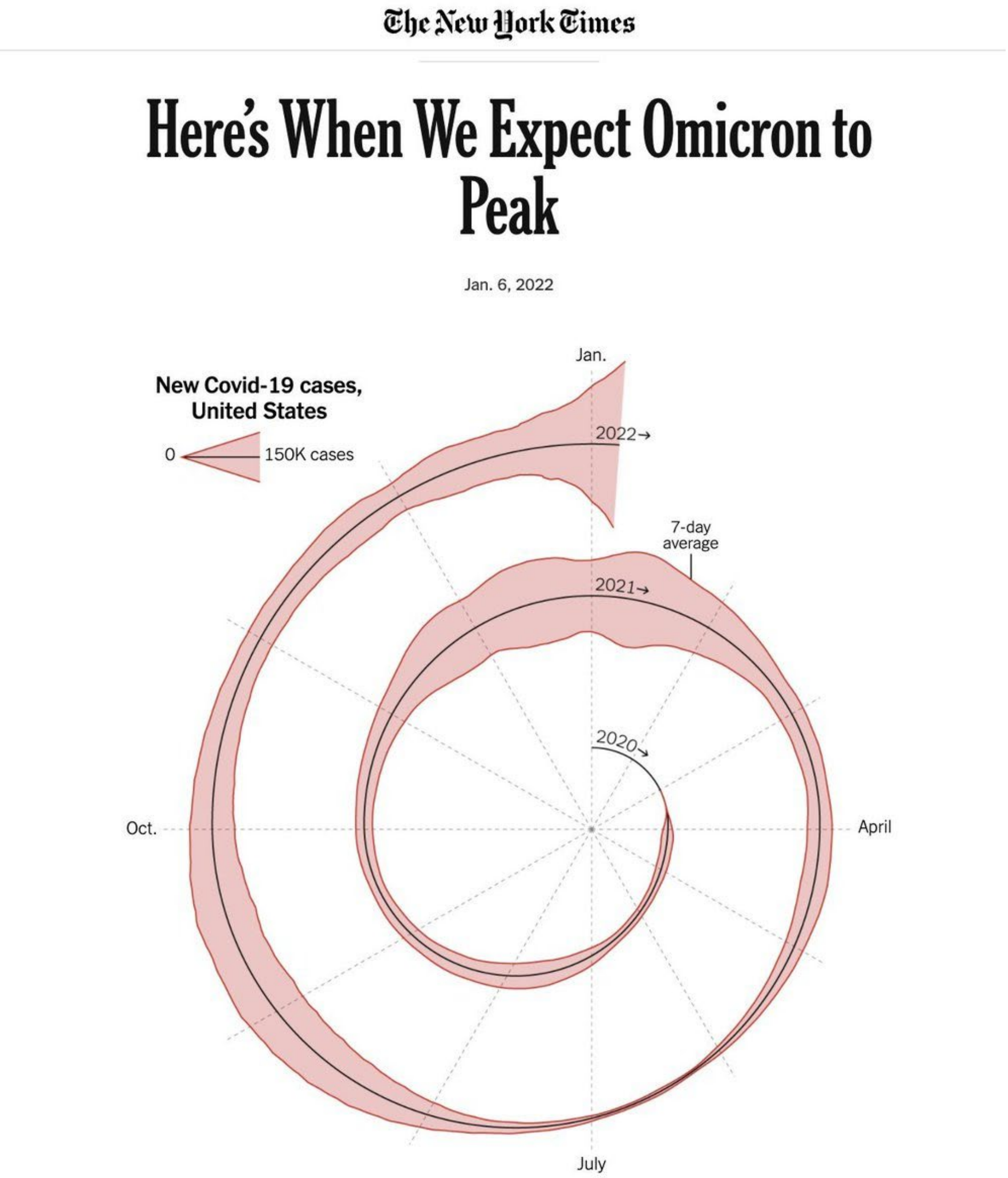
1. Truncating the y-axis
2. Univariate data portrayed with multiple dimensions
3. Design elements

Link

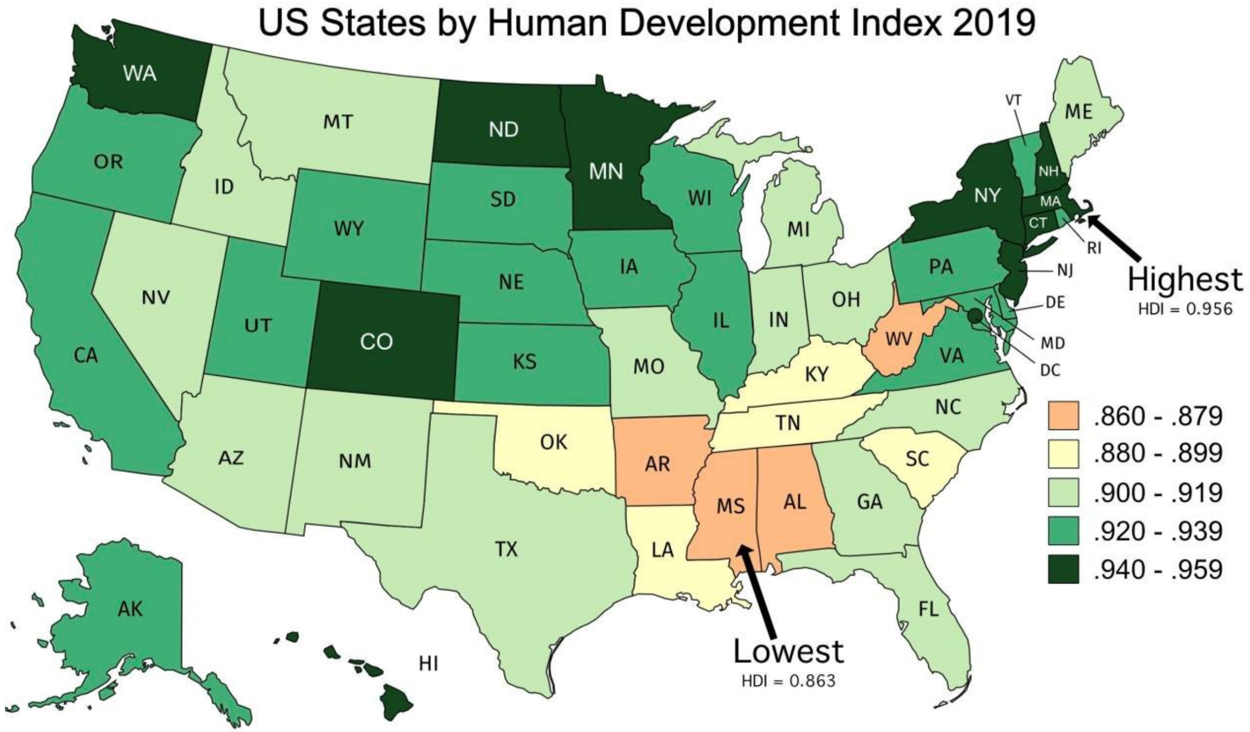




Where is the violation here?



Another principle: show data in context



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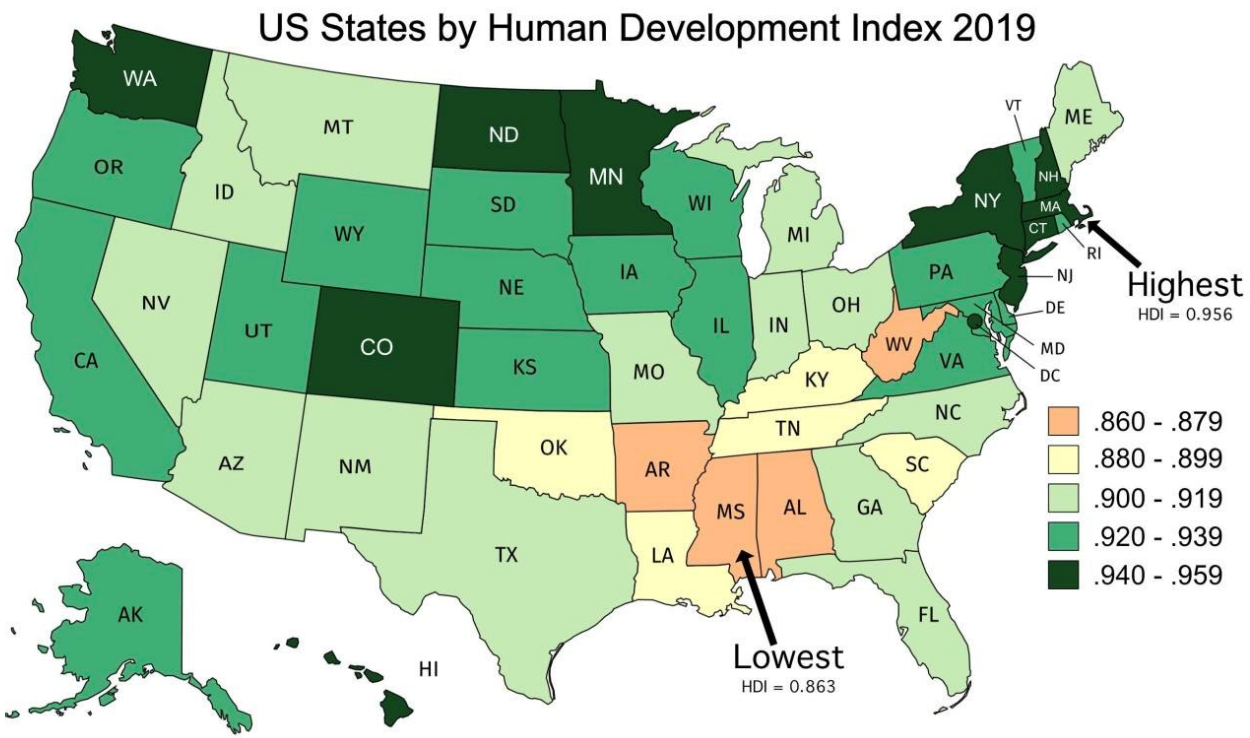
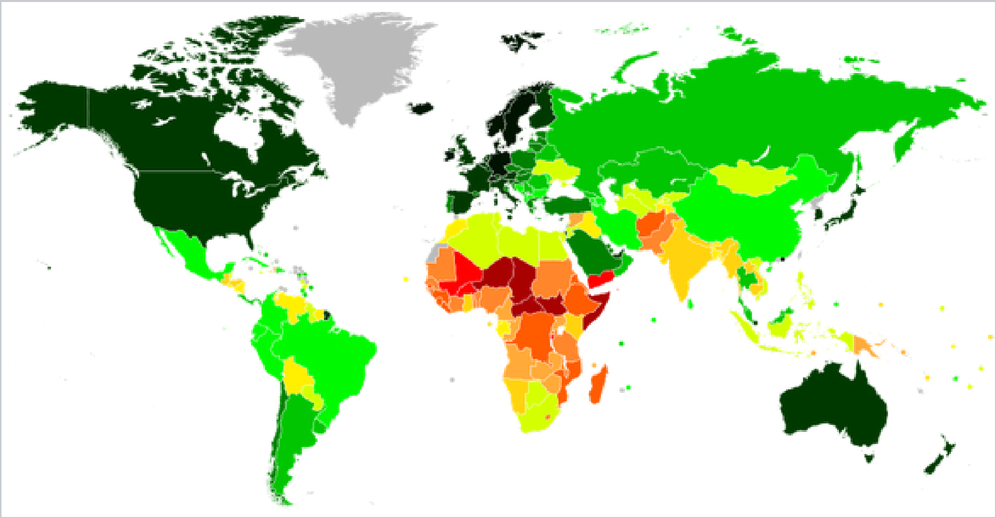




World map of countries and territories by HDI scores in increments of 0.050 (based on 2022 data, published in 2024)

≥ 0.950	0.700–0.749	0.450–0.499
0.900–0.950	0.650–0.699	0.400–0.449
0.850–0.899	0.600–0.649	≤ 0.399
0.800–0.849	0.550–0.599	Data unavailable
0.750–0.799	0.500–0.549	

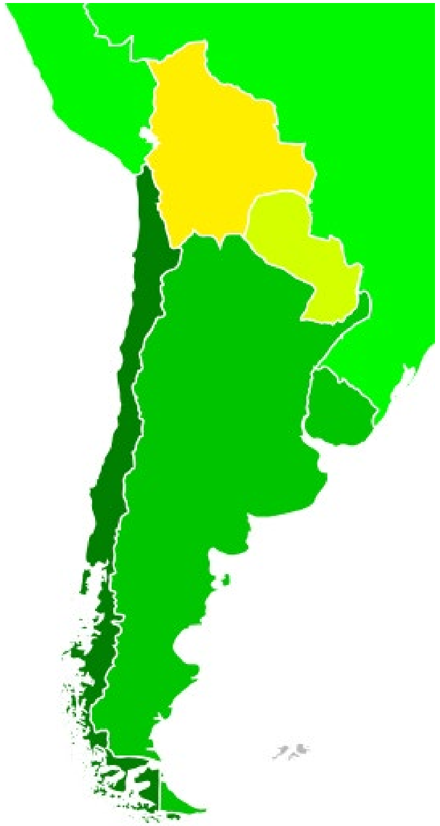
Another principle: show data in context

Tufte points to one way to deceive people, which is to not give proper context to data.

San Marino	0.867
Chile	0.860

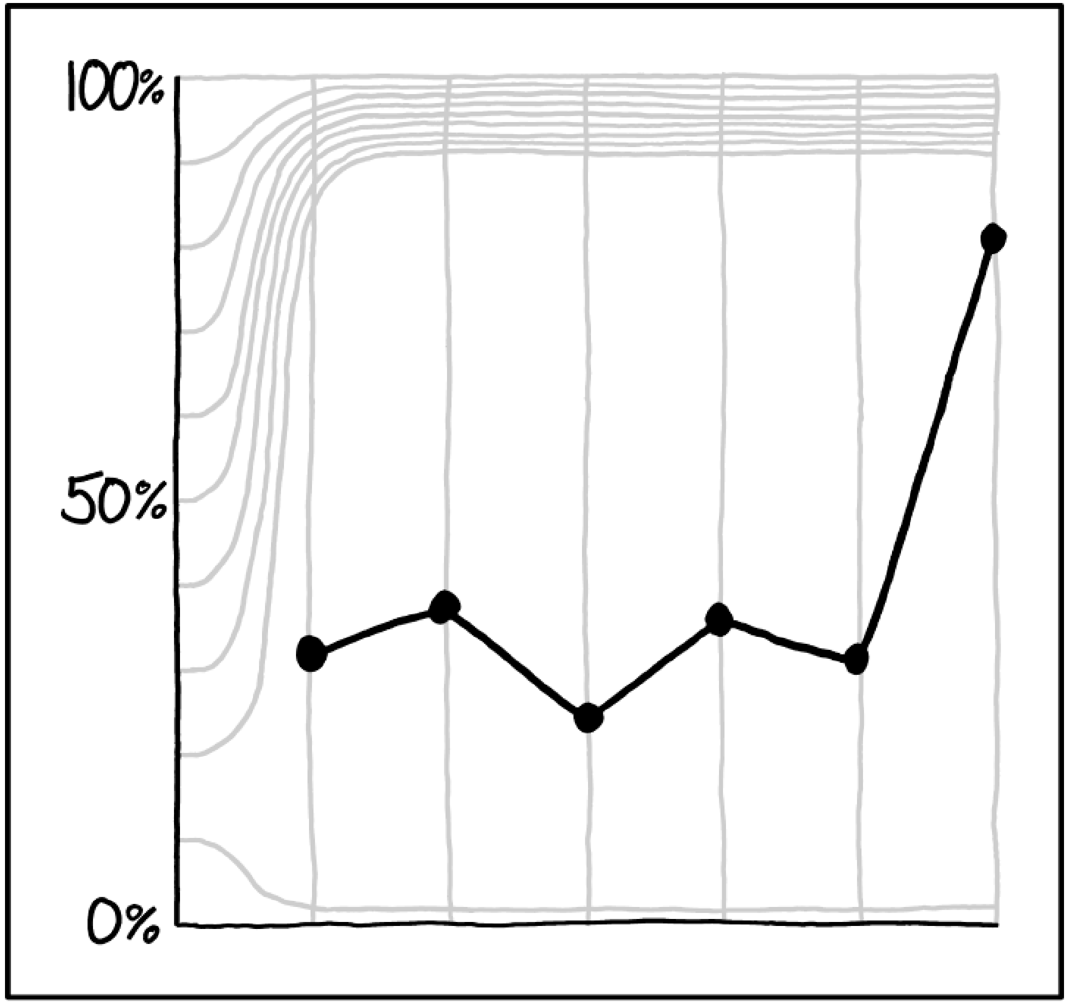
In the HDI map of the US, four states are singled out to be red, like in the country graph…

… even though they are really somewhere between Chile and San Marino, dark green on the map.



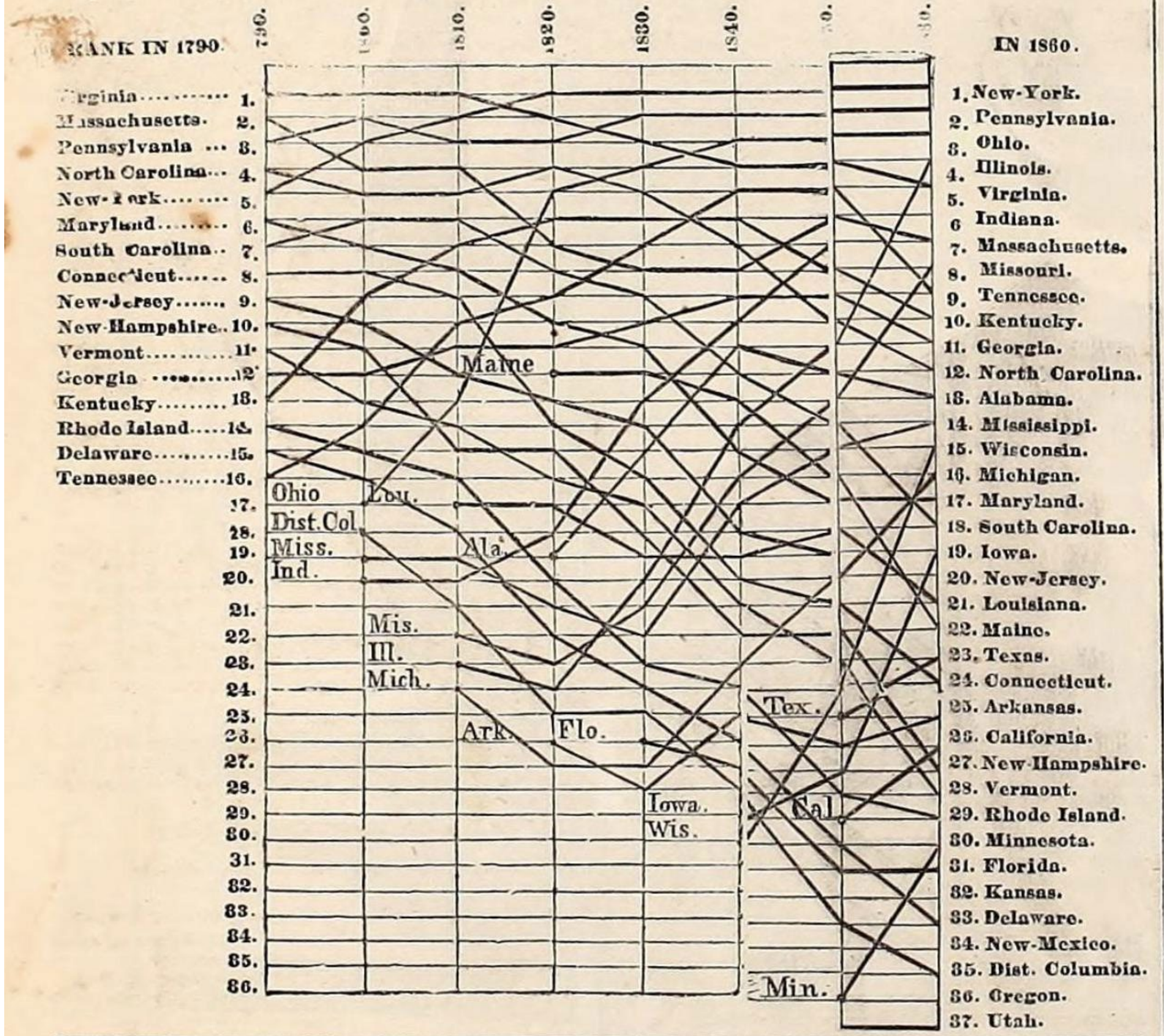
Graph case study

Using the next four graphs, discuss with your group how you might make some “Tufte-style” edits to make them clearer.

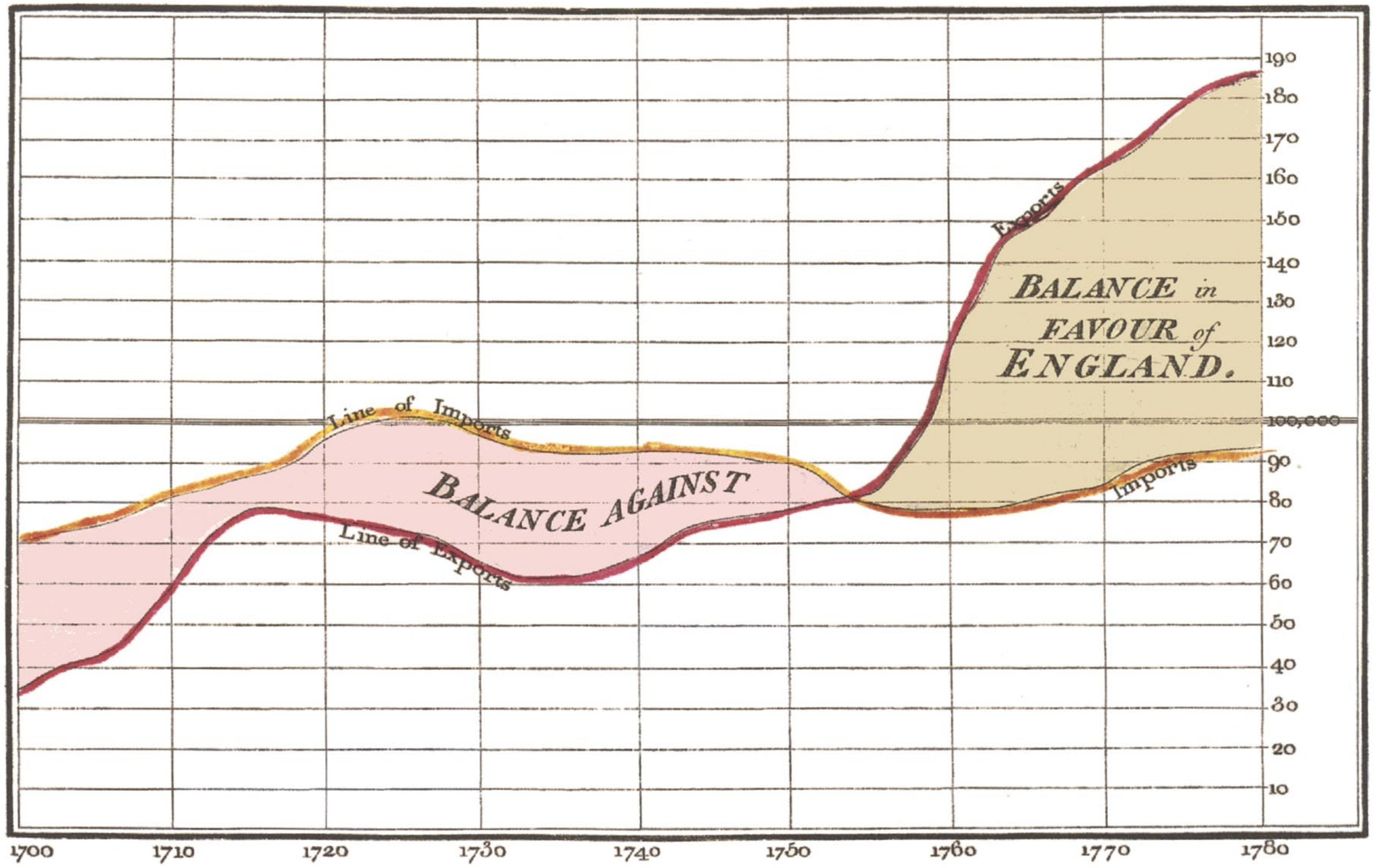


PEOPLE HAVE WISED UP TO THE "CAREFULLY CHOSEN Y-AXIS RANGE" TRICK, SO WE MISLEADING GRAPH MAKERS HAVE HAD TO GET CREATIVE.

The following Table, for which we are indebted to the courtesy of the New York *Times*, shows the progress of population in each State from 1790 to 1860:



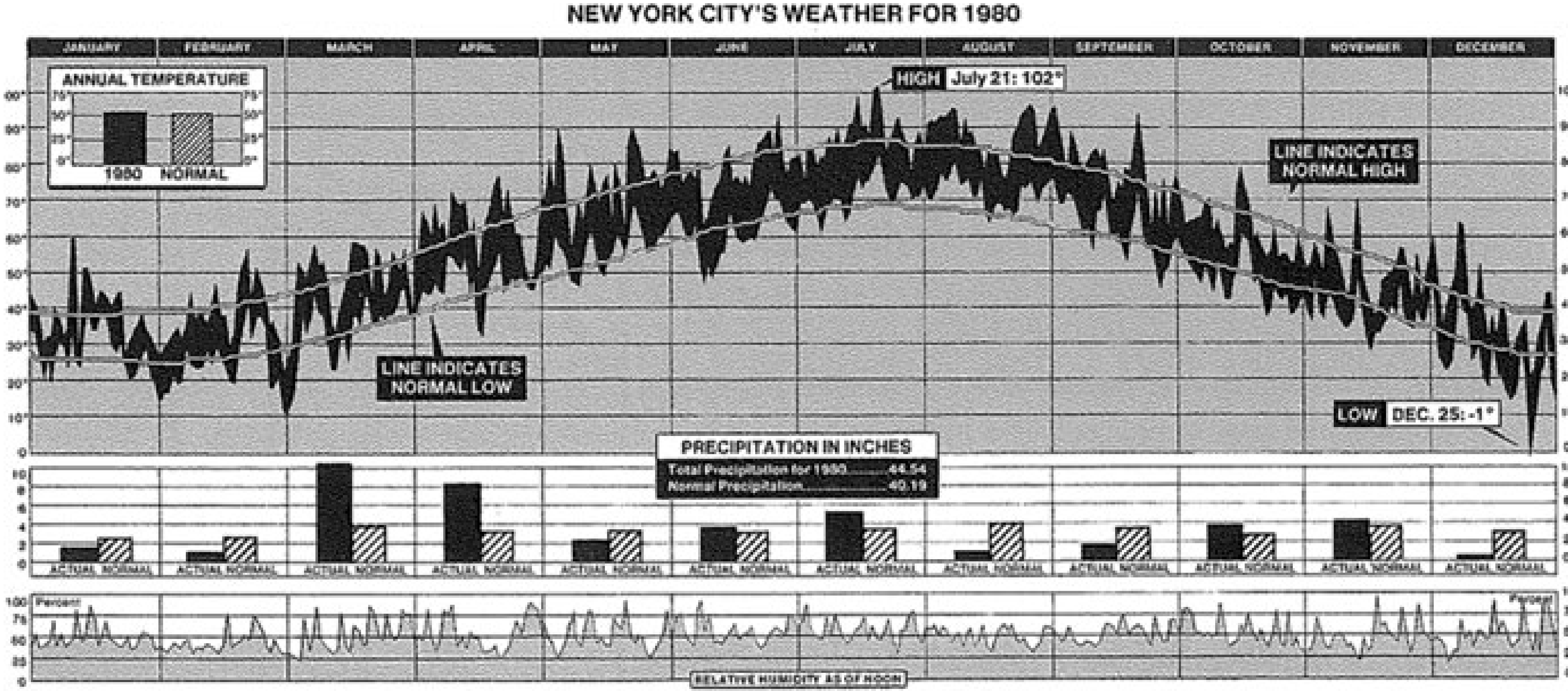
Exports and Imports to and from DENMARK & NORWAY from 1700 to 1780.



The Bottom line is divided into Years, the Right hand line into £10,000 each.

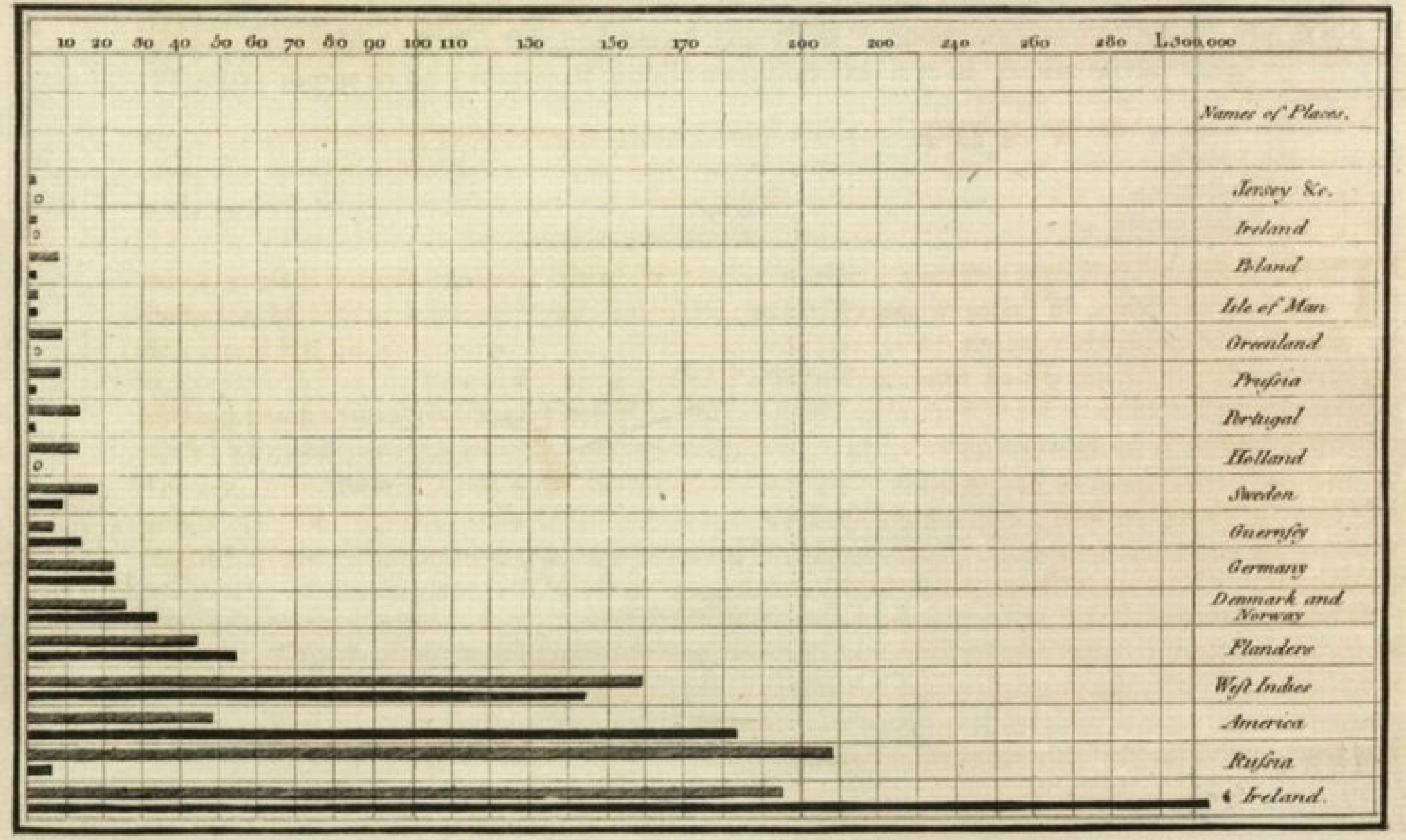
Published as the Act directs, 14th May 1786, by Wm. Playfair

Neele sculpt 352, Strand, London.

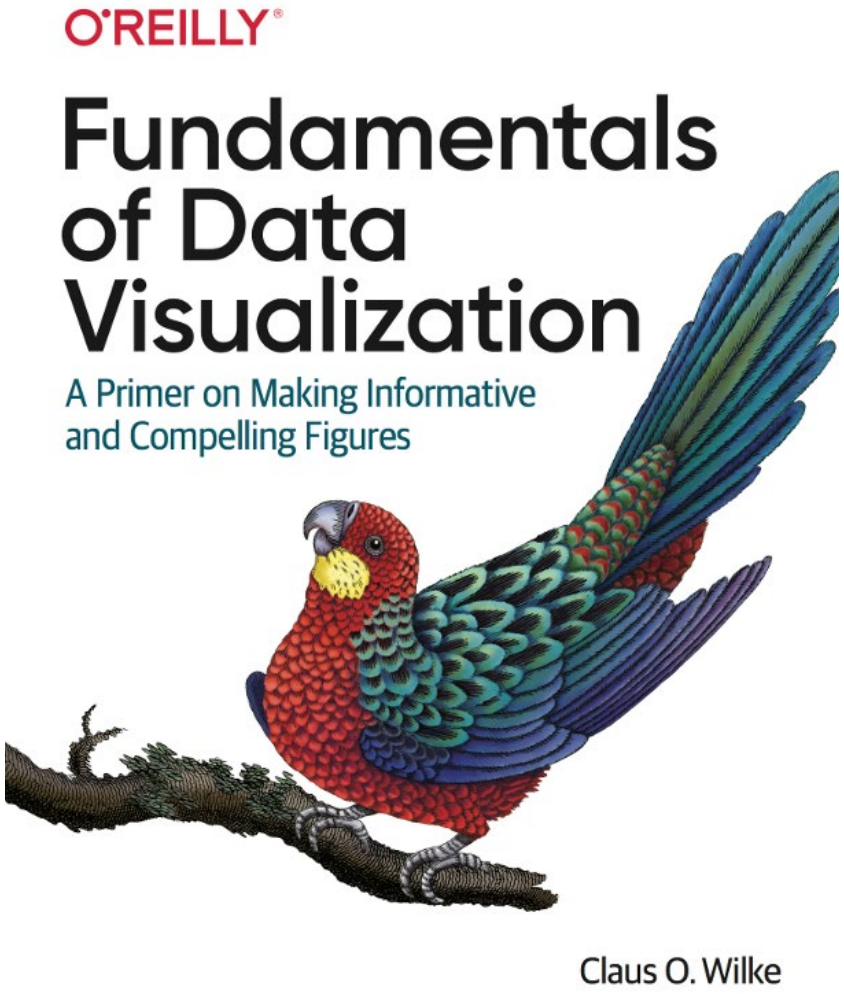


New York Times, January 11, 1981, p. 32.

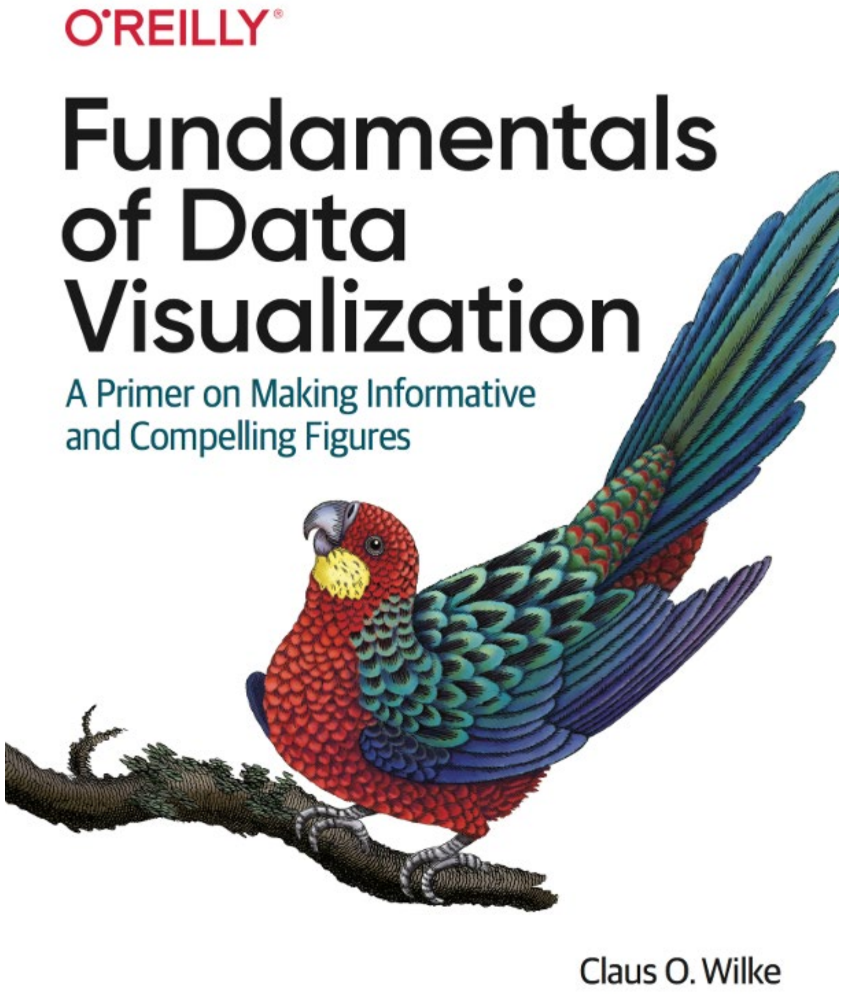
Exports and Imports of SCOTLAND to and from different parts for one Year from Christmas 1780 to Christmas 1781.



The Upright divisions are Ten Thousand Pounds each. The Black Lines are Exports the Ribbed lines Imports.



Data Visualization Principles: Lessons on Color from Wilke



Claus Wilke's book is a free text that has some particularly good guidance about using color.

Wilke mentions three ways to use color:

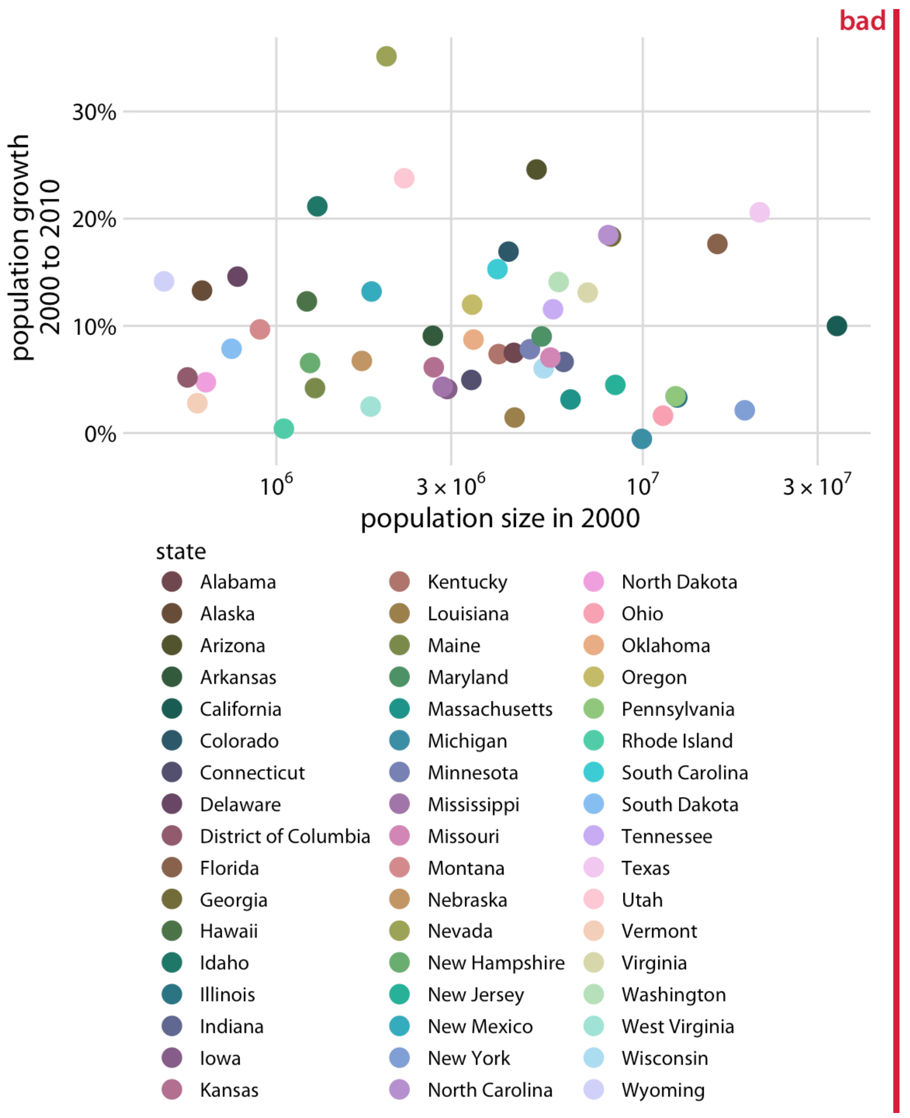
1. Color as a tool to distinguish
2. Color to represent data values
3. Color as a tool to highlight

Color to distinguish

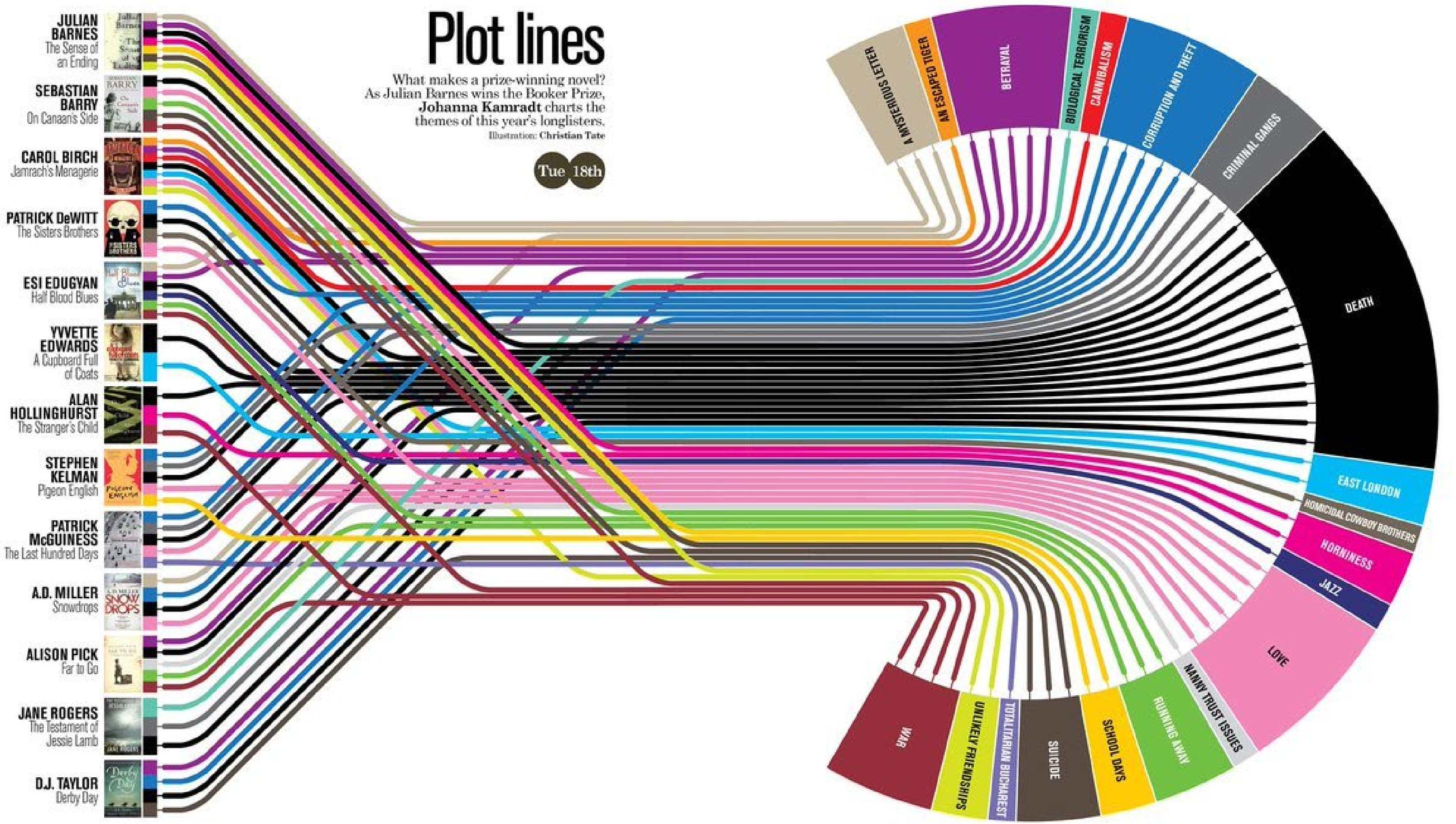
Color used this way is for discrete values and categories, not continuous variables.

Rule of thumb: people ***cannot distinguish more than 5 categories*** with color.

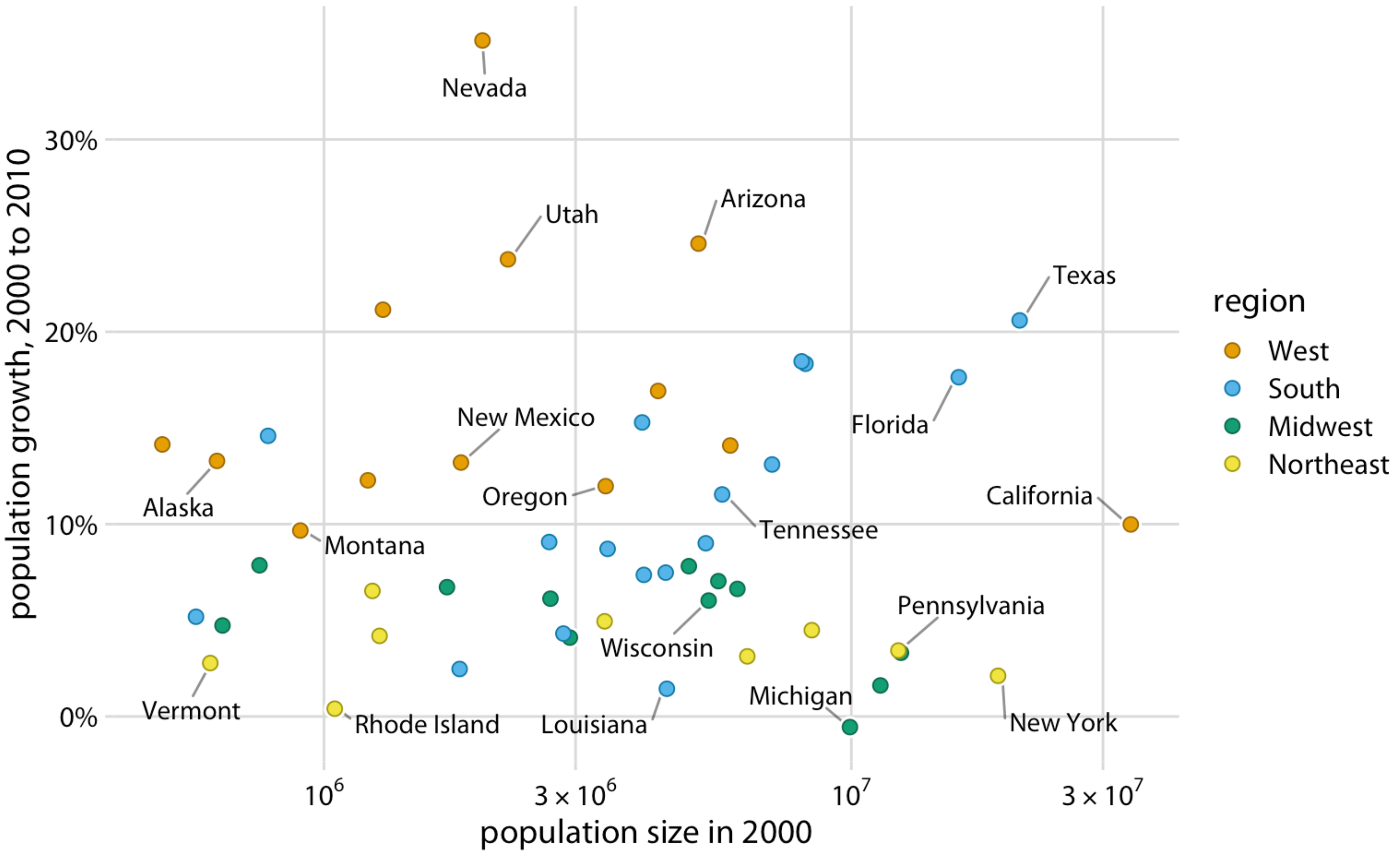
(I see a shocking number of graphs at the PhD level that can't get this right).



Color to distinguish



Color to distinguish

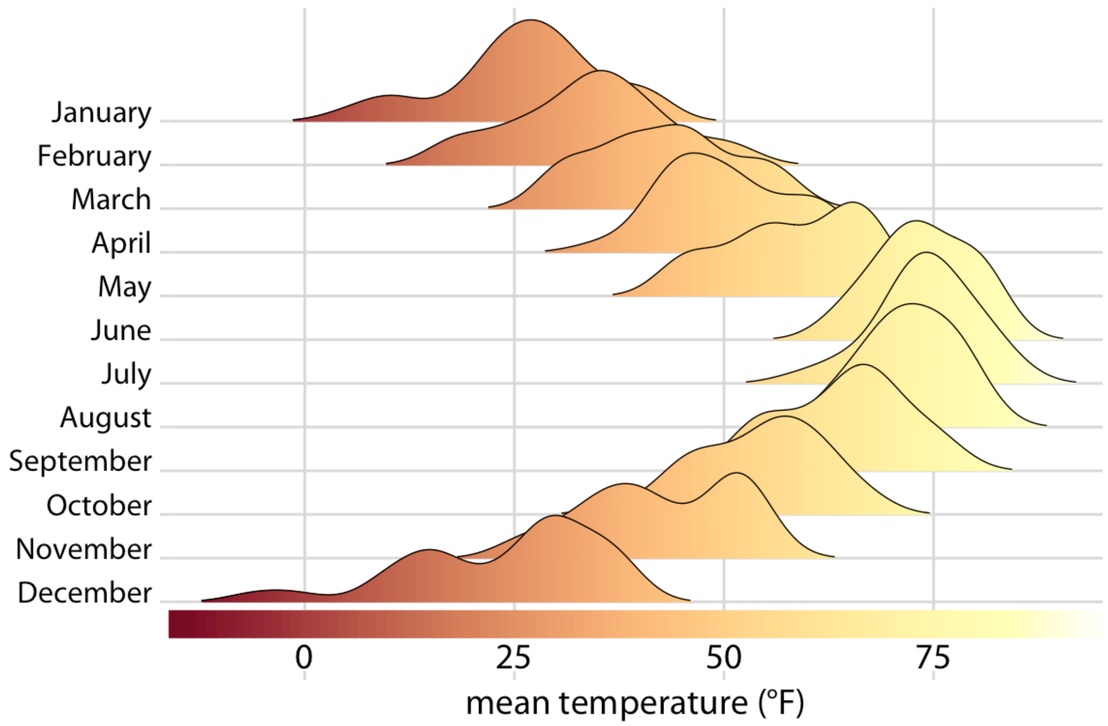


Color to represent data values

To represent a continuous variable, most visualizations use a colormap.

If the variable is ***continuous,*** the colormap must be ***monotonic.***

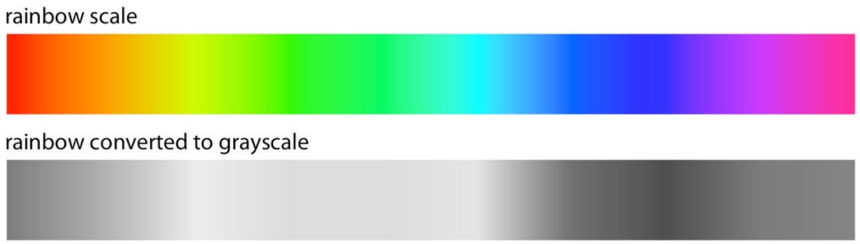
In other words, changes in shade have to be uniform if changes are uniform.

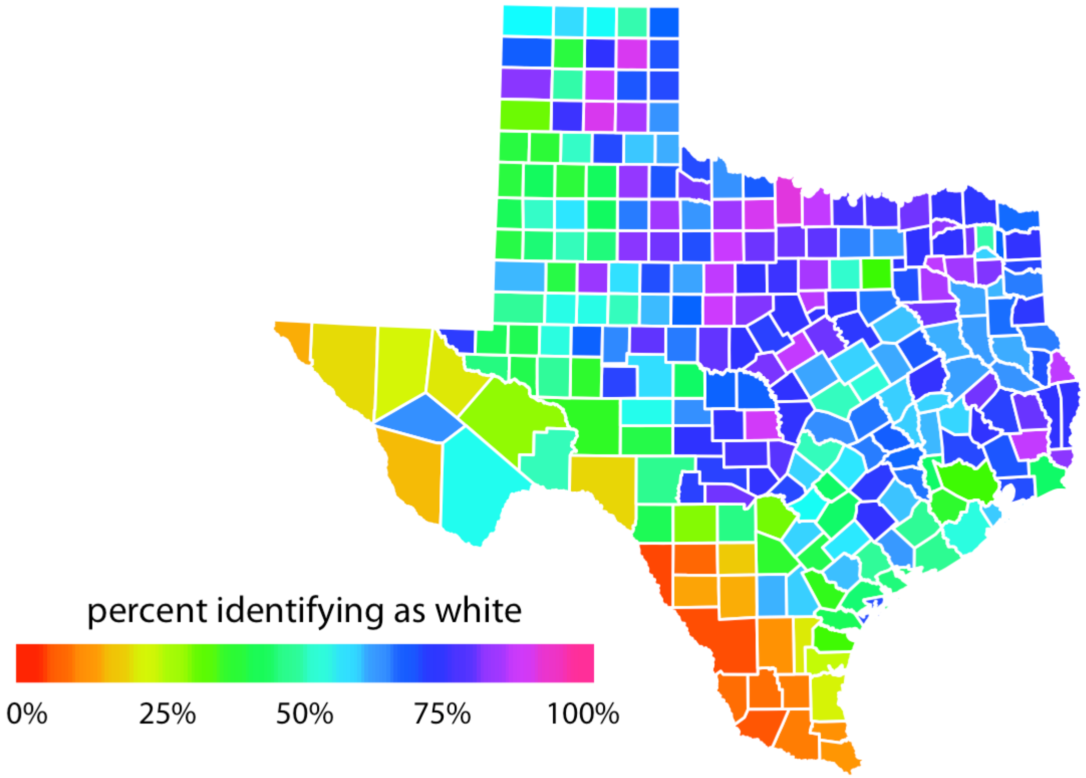


Color to represent data values

The rainbow colormap (often called "Jet") is the most famous culprit.

The changes in the rainbow when mapped to gray clearly don't change consistently with the variable.



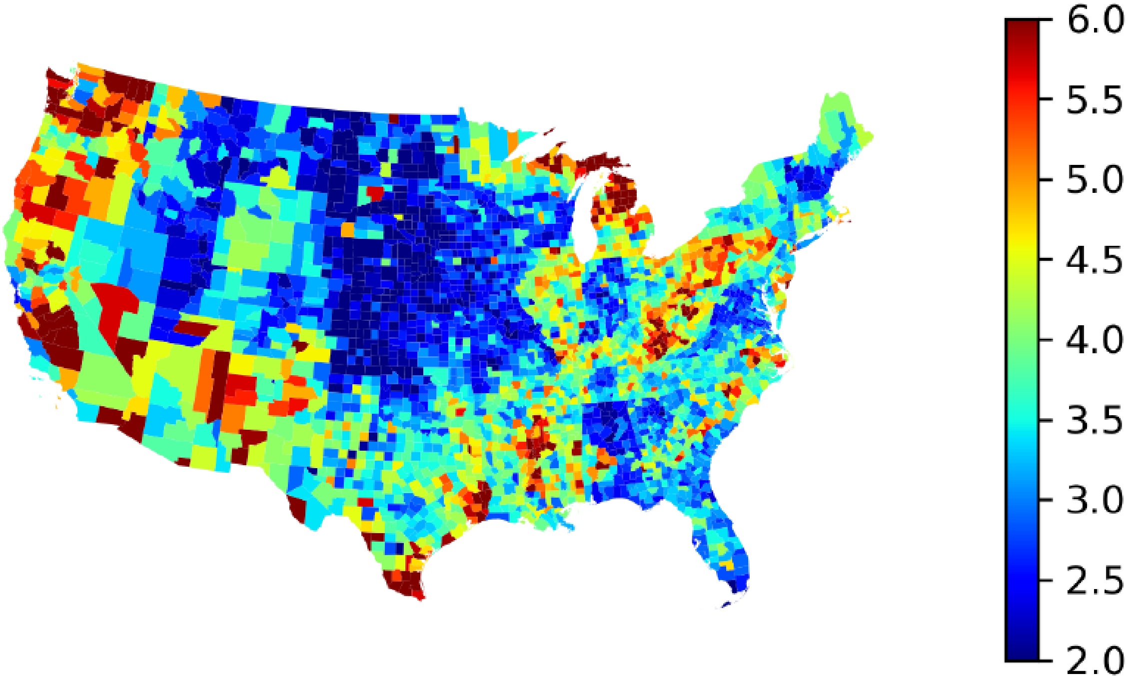


bad

Color to represent data values

What categories are your eyes making when you look at this colormap?

Figure 5: County Unemployment Rates in 2022



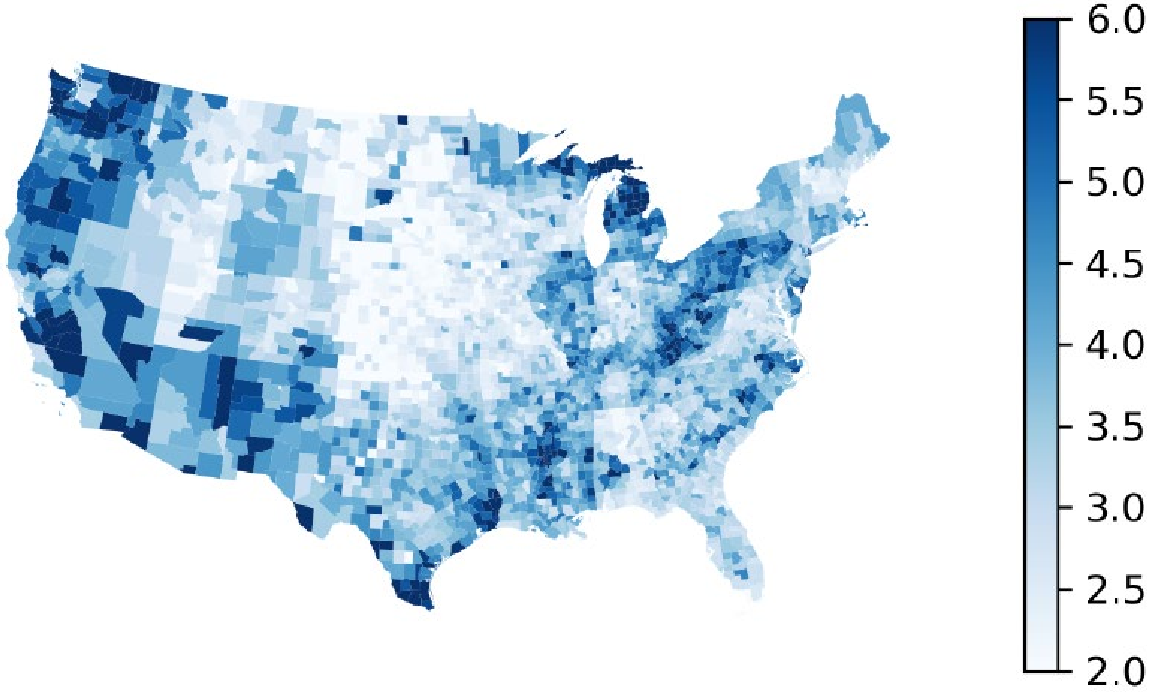
(a) Non-Monotonic Colormap ("Jet")

Color to represent data values

What categories are your eyes making when you look at this colormap?

How about now?

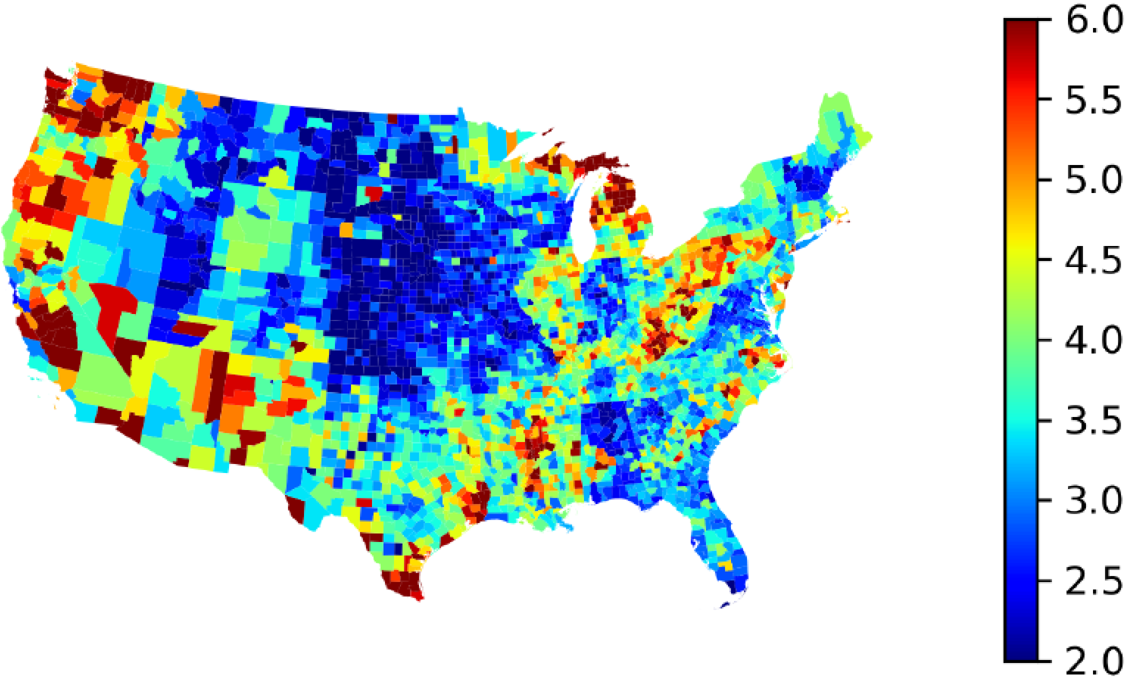
Figure 5: County Unemployment Rates in 2022



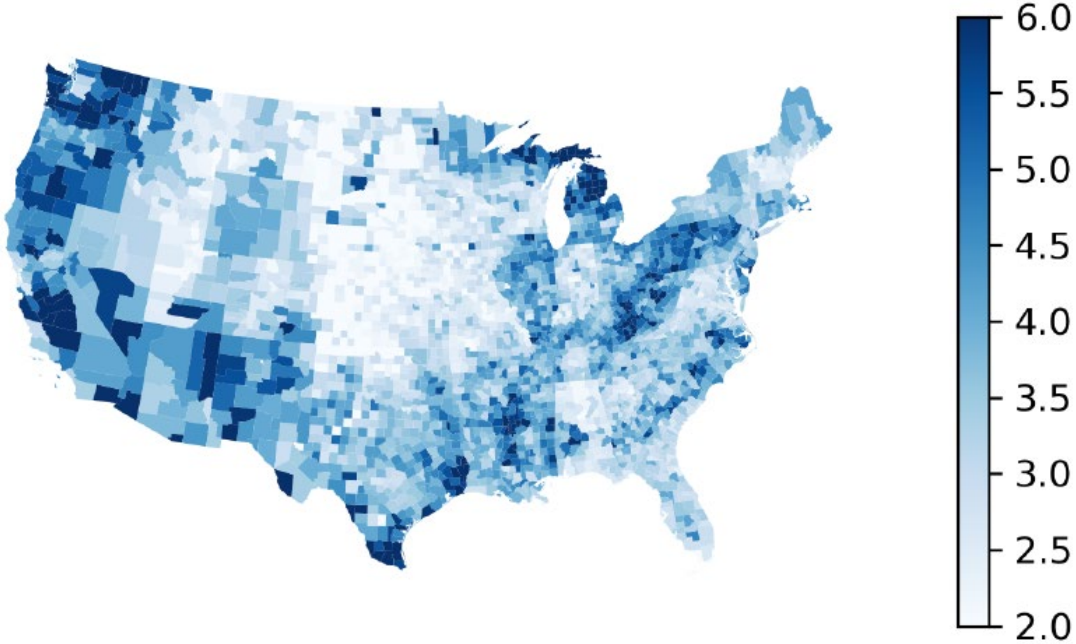
(b) Monotonic Colormap ("Blues")

Color to represent data values

Figure 5: *County Unemployment Rates in 2022*



(a) Non-Monotonic Colormap ("Jet")



(b) Monotonic Colormap ("Blues")

Color to represent data values

Color maps like the ones here are best for discrete categories, not continuous data

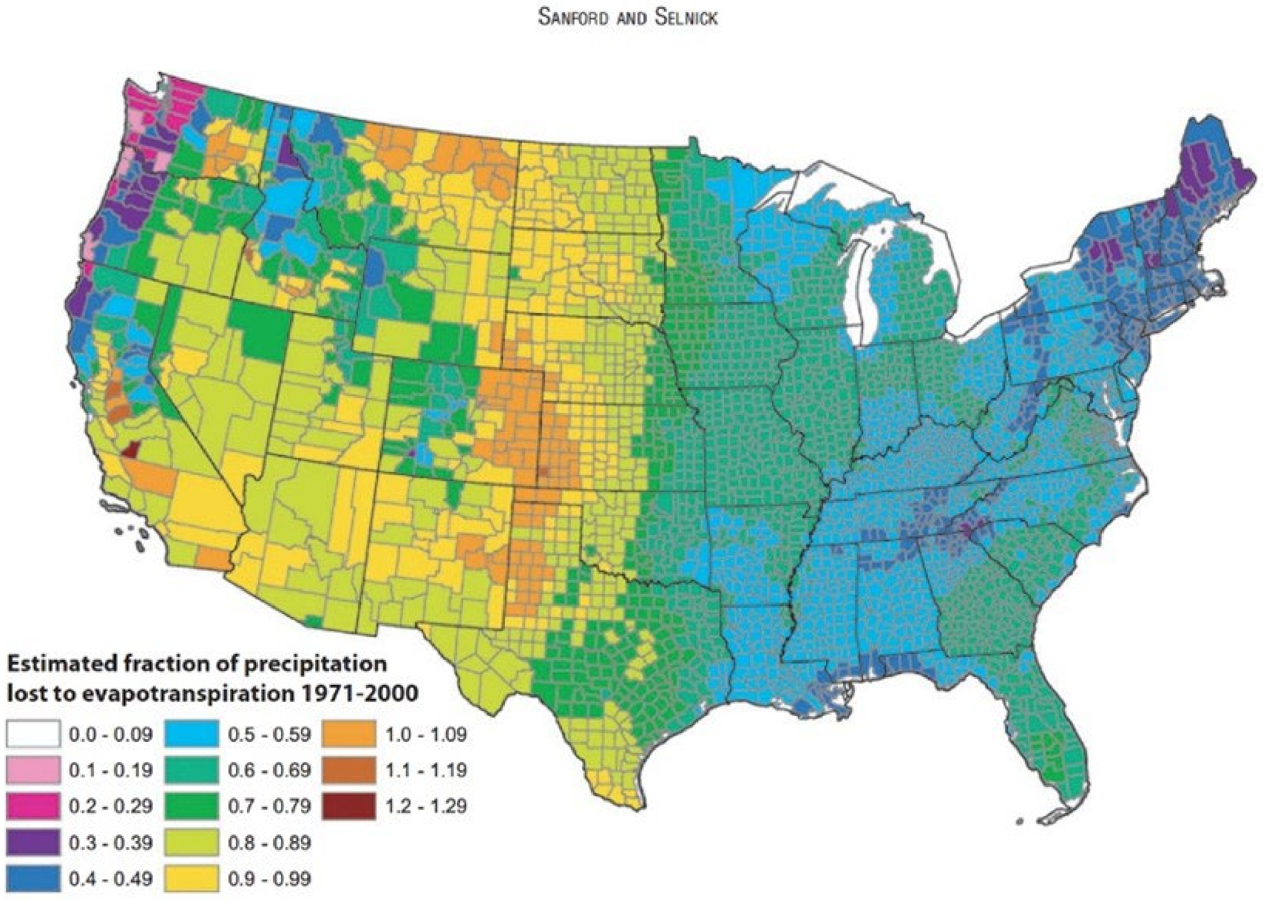


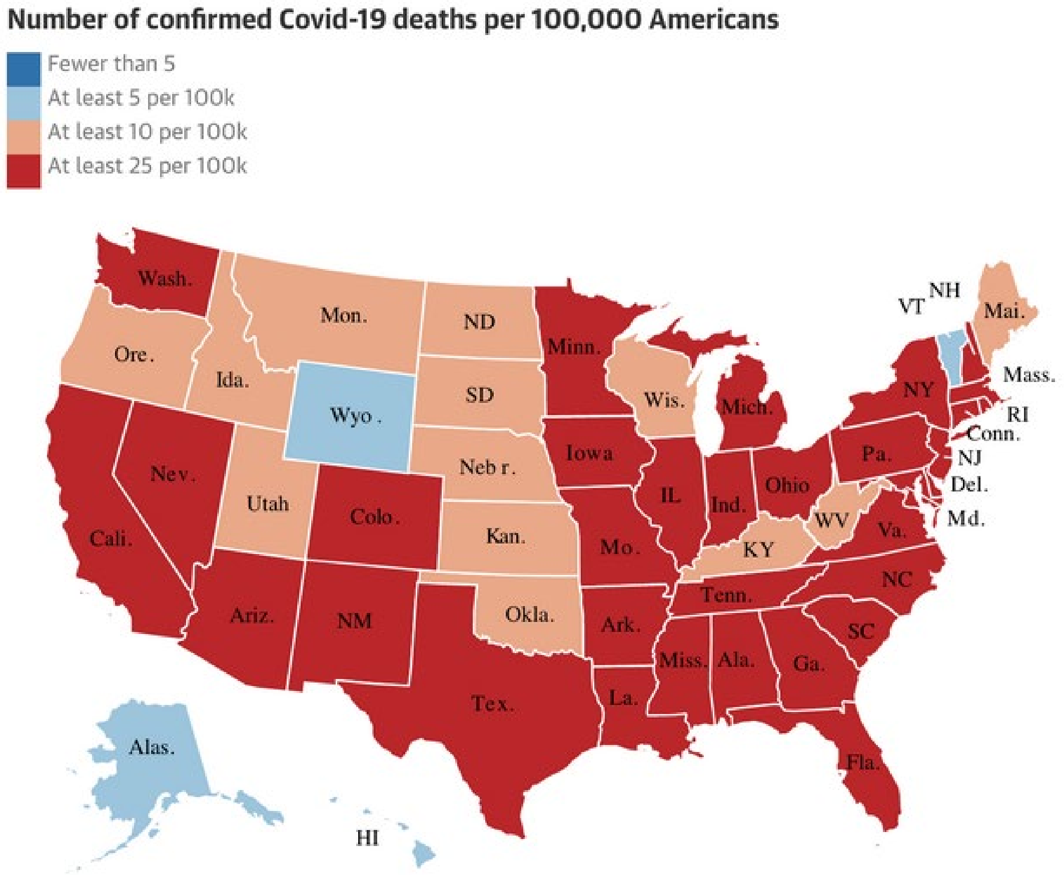
FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

Another colormap pitfall

I see a large number of graphs that use diverging colormaps with diverging points that make no sense, either because:

- The data are not diverging
- The diverging point is random

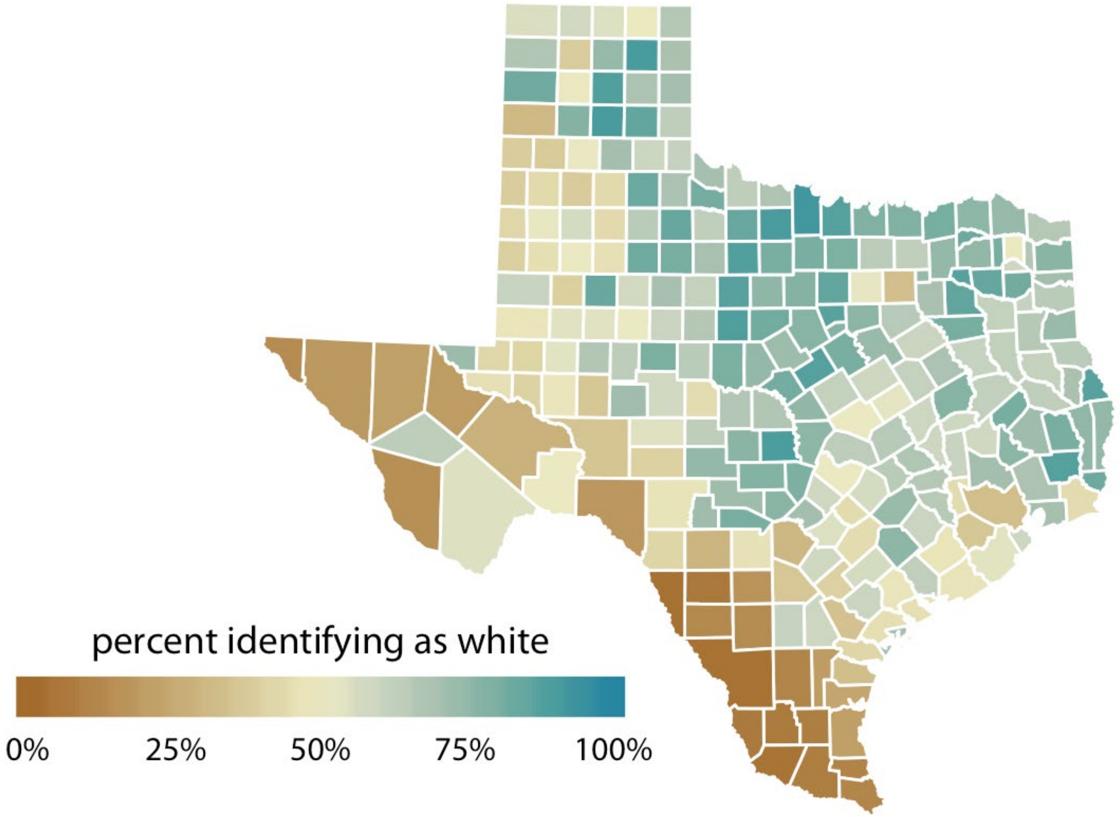
Red-Blue is a popular choice, but the graph on the right does not need to be diverging.



Another colormap pitfall

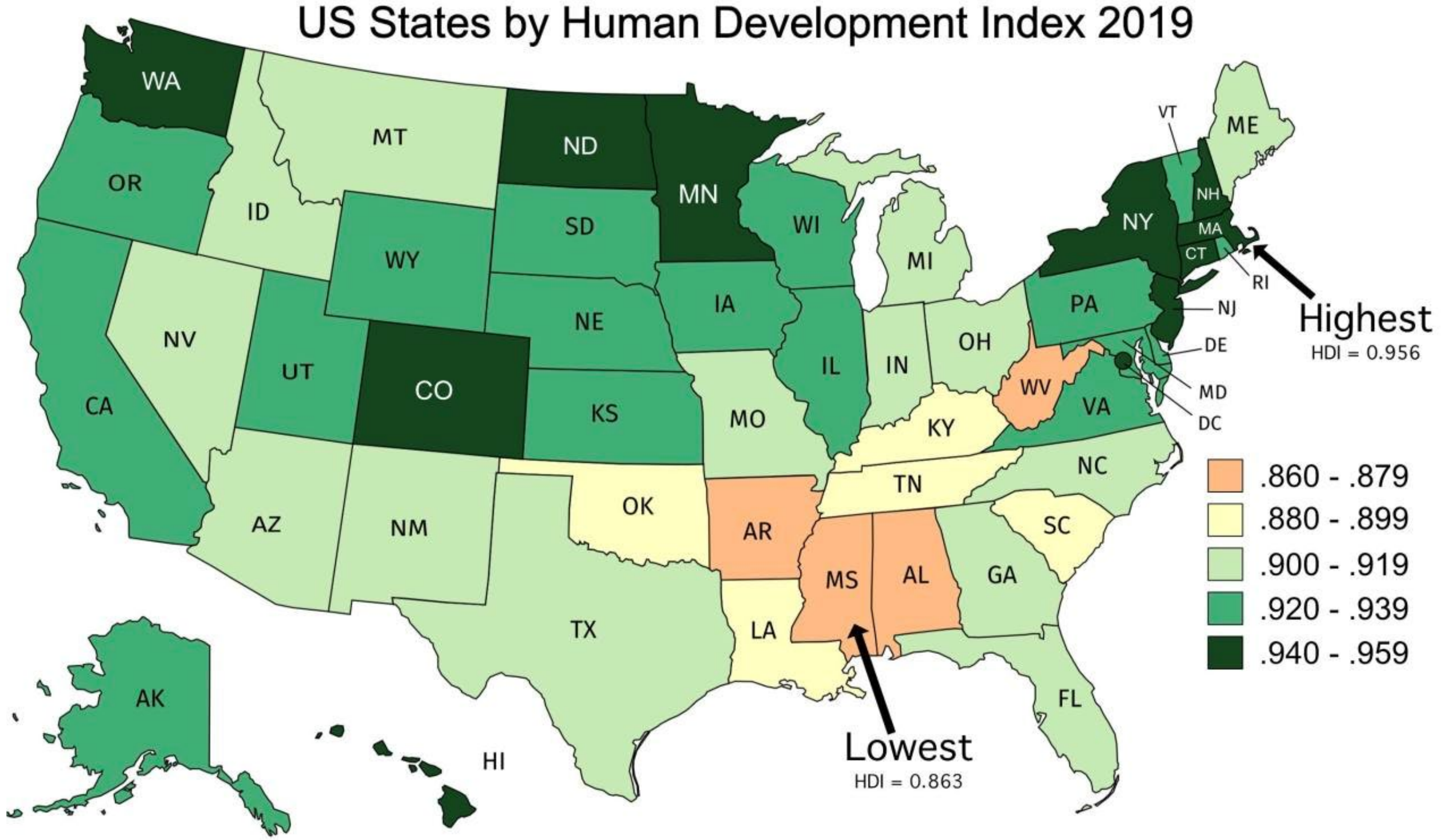
In fact, making it a diverging colormap completely changes the interpretation.

What is the value being highlighted in the graph on the right?



Another colormap pitfall

How about here?

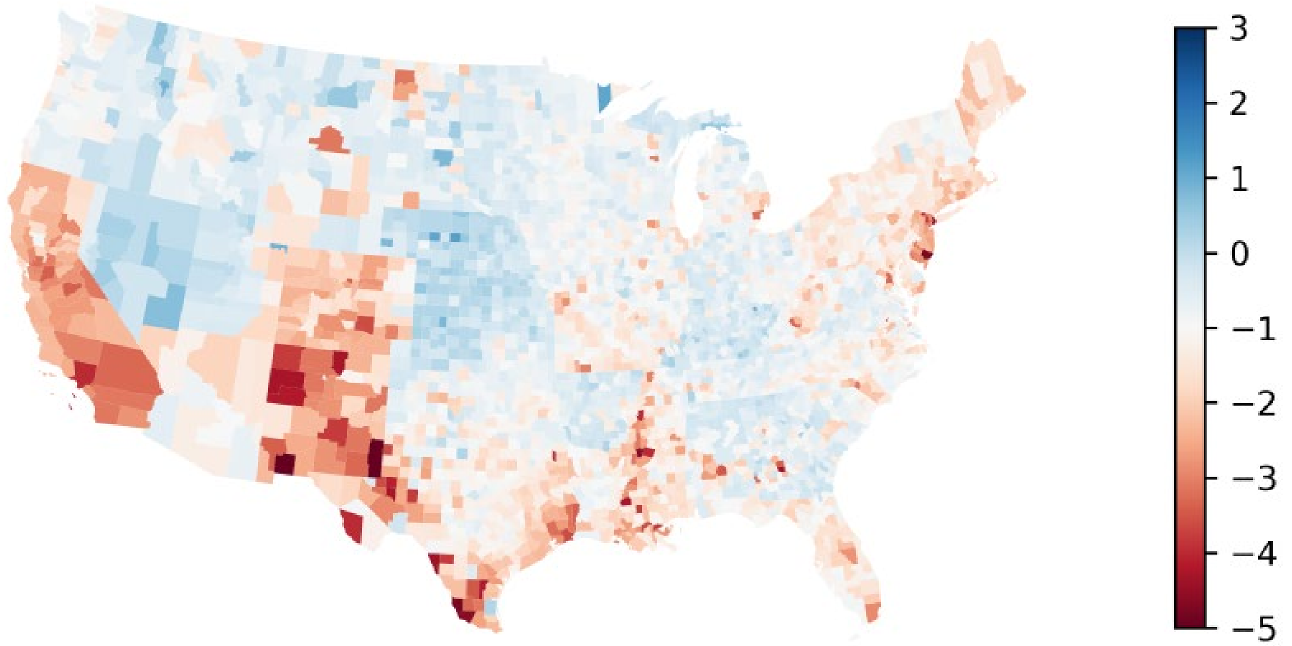


Another colormap pitfall

Another common mistake: an arbitrary diverging point.

What do you conclude about the graph on the right?

Figure 6: *Change in County Unemployment Rates between 2021 and 2022*

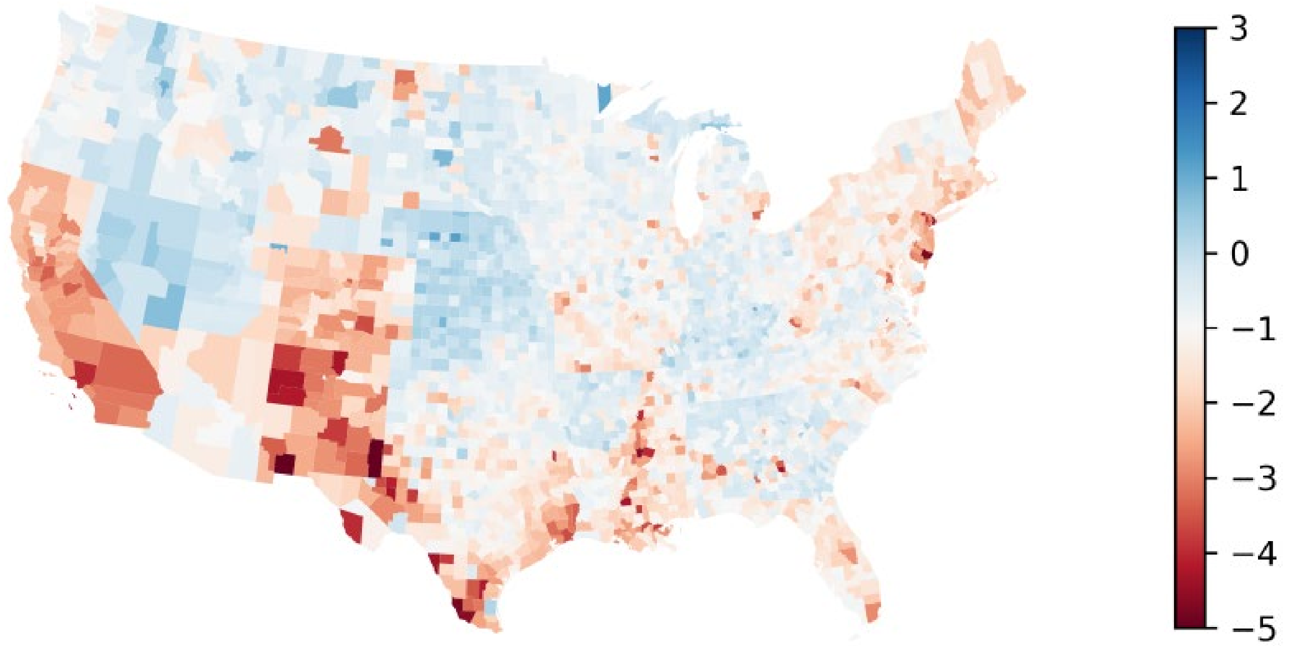


Another colormap pitfall

Using -1 as the diverging point colored made roughly half the counties red and half the counties blue.

What if we made 0 as the diverging point?

Figure 6: *Change in County Unemployment Rates between 2021 and 2022*



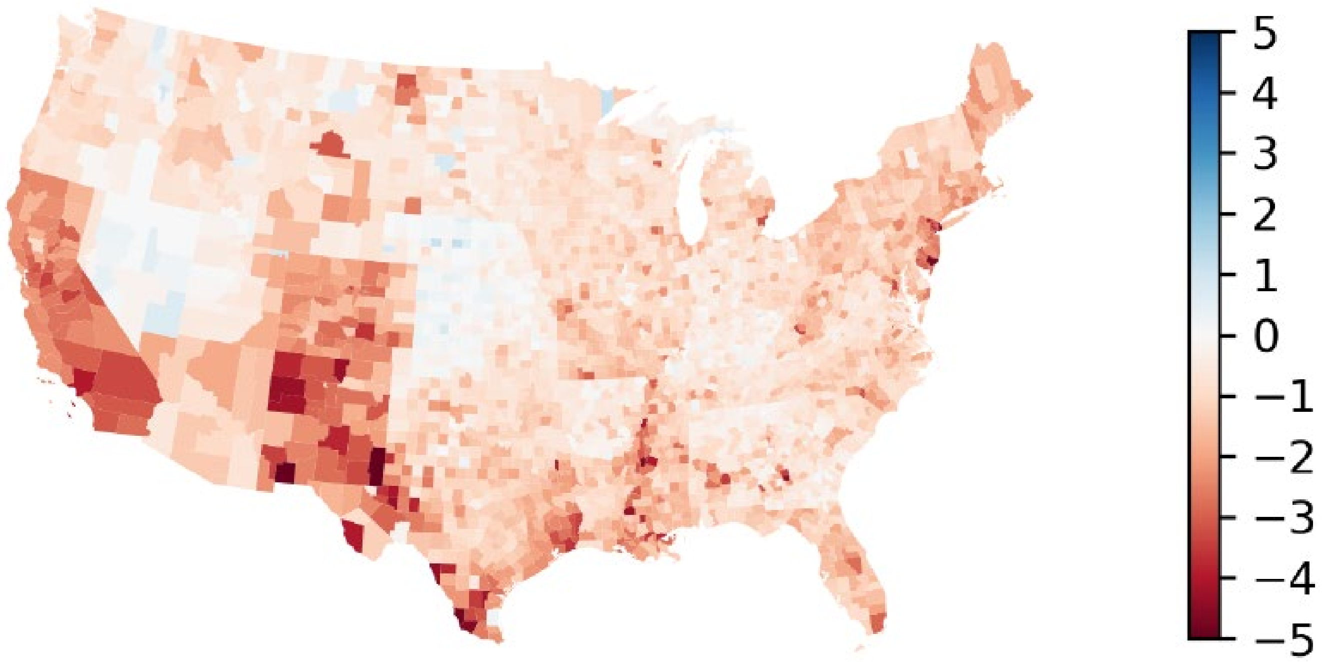
Another colormap pitfall

Using -1 as the diverging point colored made roughly half the counties red and half the counties blue.

What if we made 0 as the diverging point?

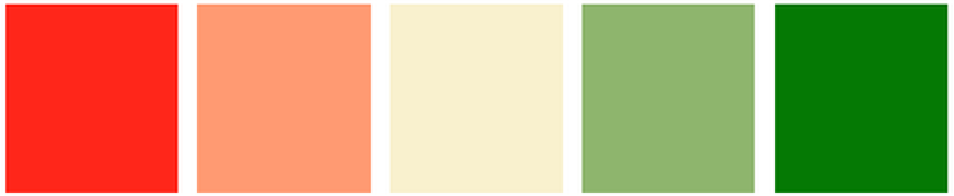
In reality, 98% of counties saw a drop in unemployment.

Figure 6: Change in County Unemployment Rates between 2021 and 2022



One more colormap pitfall

What's the problem with the colormap to the right?



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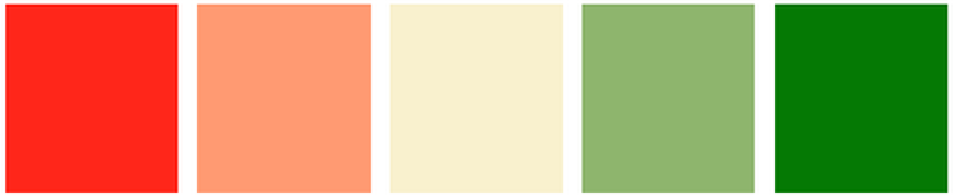

This is what it looks like to people that are R/G colorblind.

One more colormap pitfall

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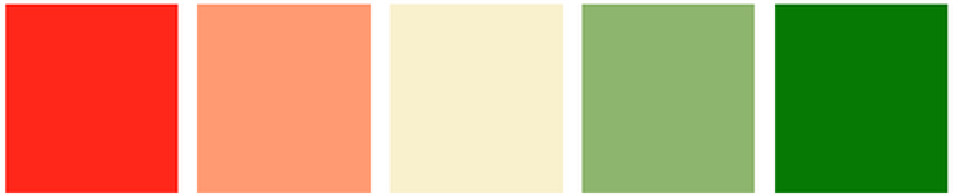


This is what it looks like to people that are R/G colorblind.

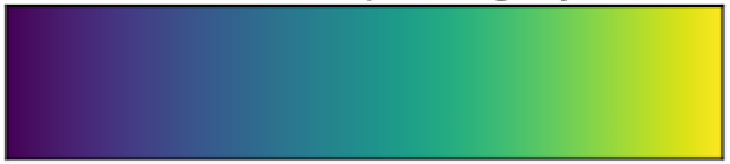
Similarly, this is what blue green can look like to some people:

One more colormap pitfall

In most graphing packages, the default is something that can be seen with any visual deficiency.

In matplotlib, the default is “viridis” which was developed by python programmers and works even in gray scale

The colormap in its glory

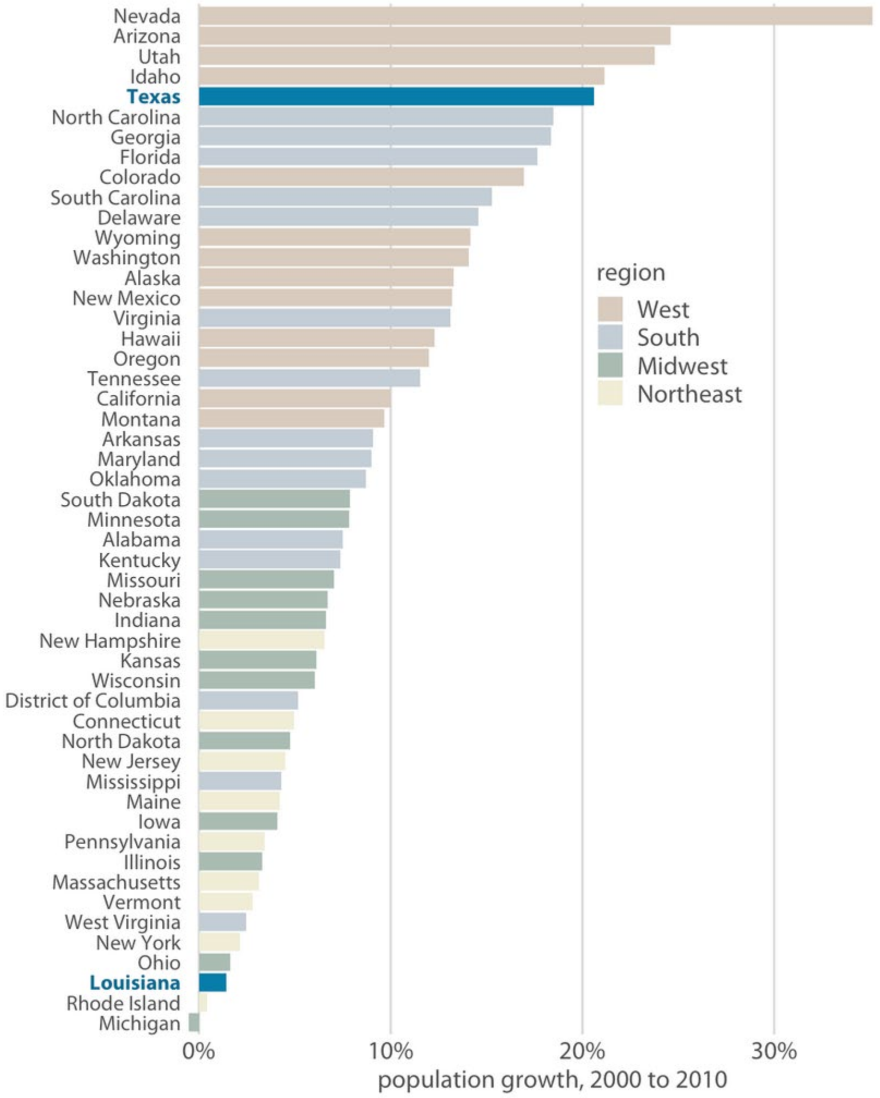


Black-and-white printed

Color to highlight

Using color sparingly is a good way of making things stick out from figures.

Sometimes this means using a colormap but saying alpha parameter low to make some colors more transparent.



Last aside about colors: intuition

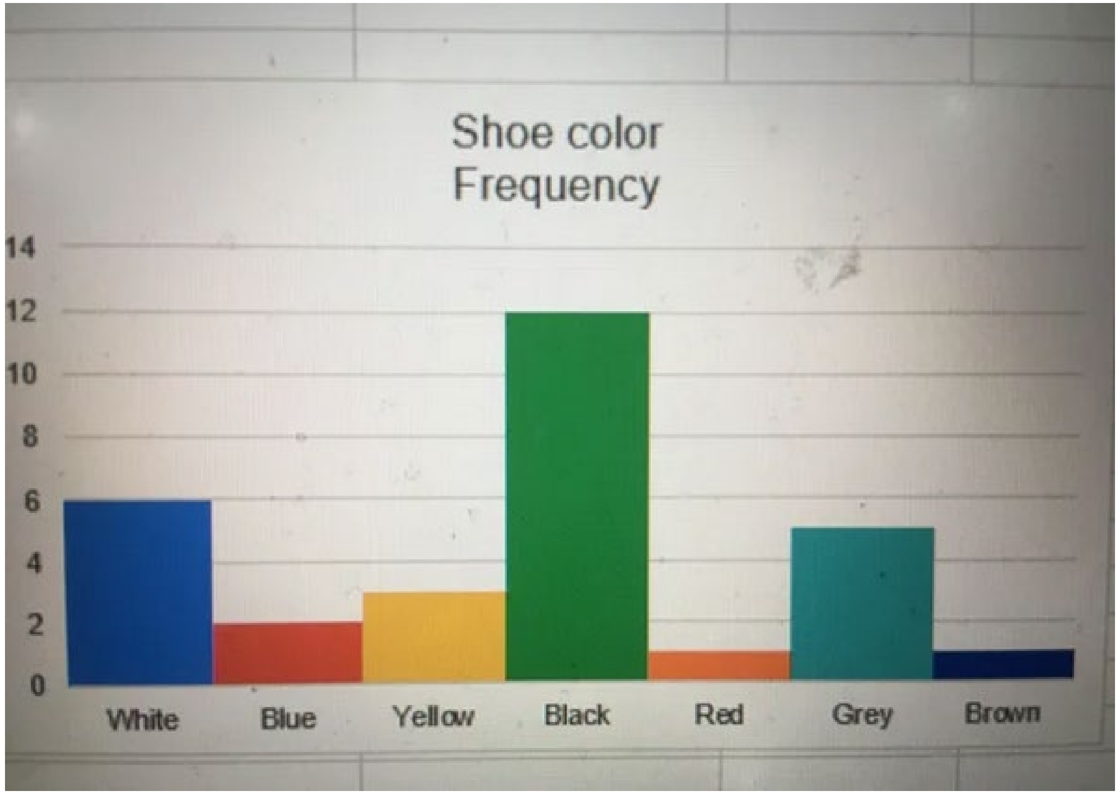
The meaning or intuition of certain colors can have a large impact on your graph.

In US context, some common ones:

- Red: negative, stop, heat
- Green: money, plants, positive
- Blue: cool, water

Colormaps tend to want to go:

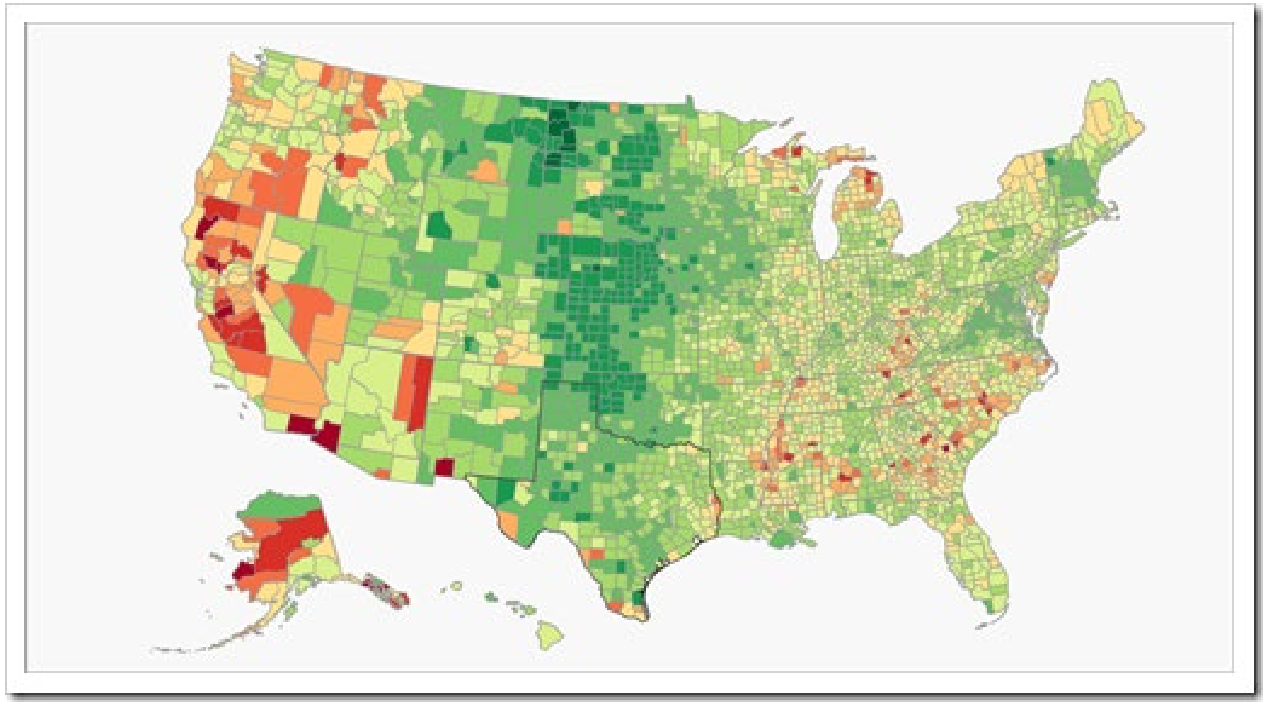
Lighter color = higher number



Last aside about colors: intuition

Culture and politics also have an impact.

When you see the figure on the right, what do you interpret the colors as?

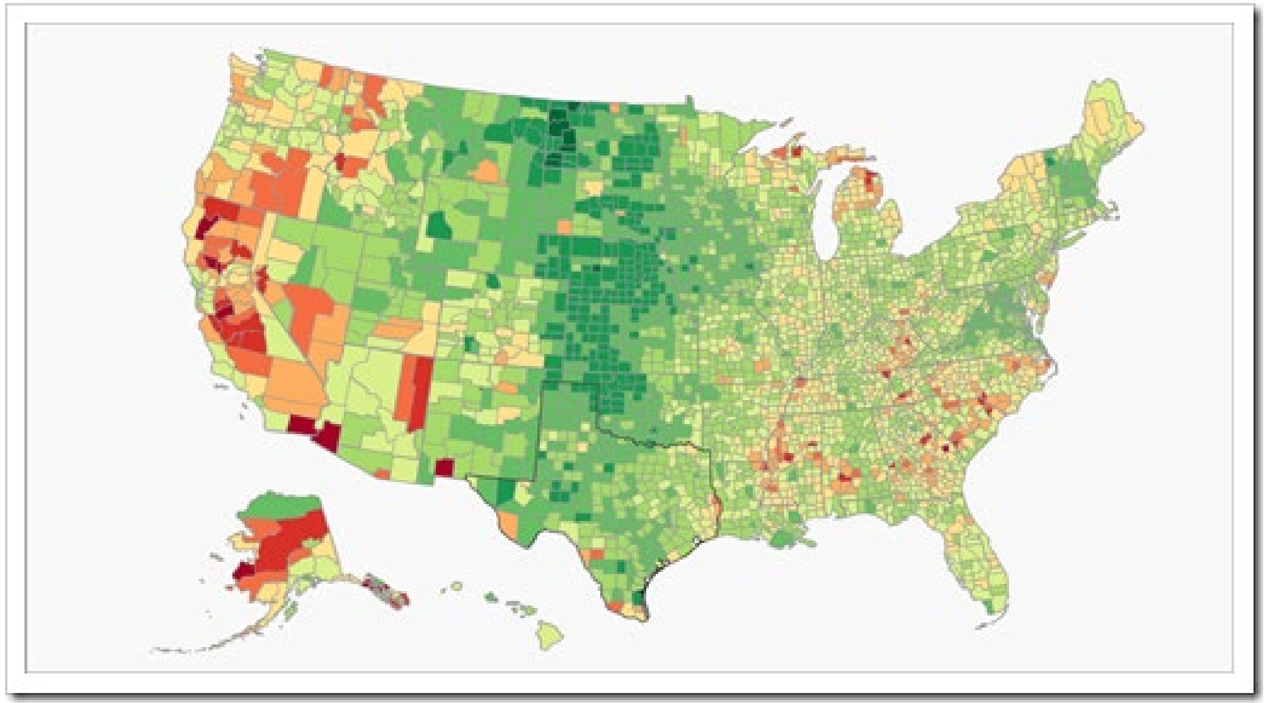


Last aside about colors: intuition

Culture and politics also have an impact.

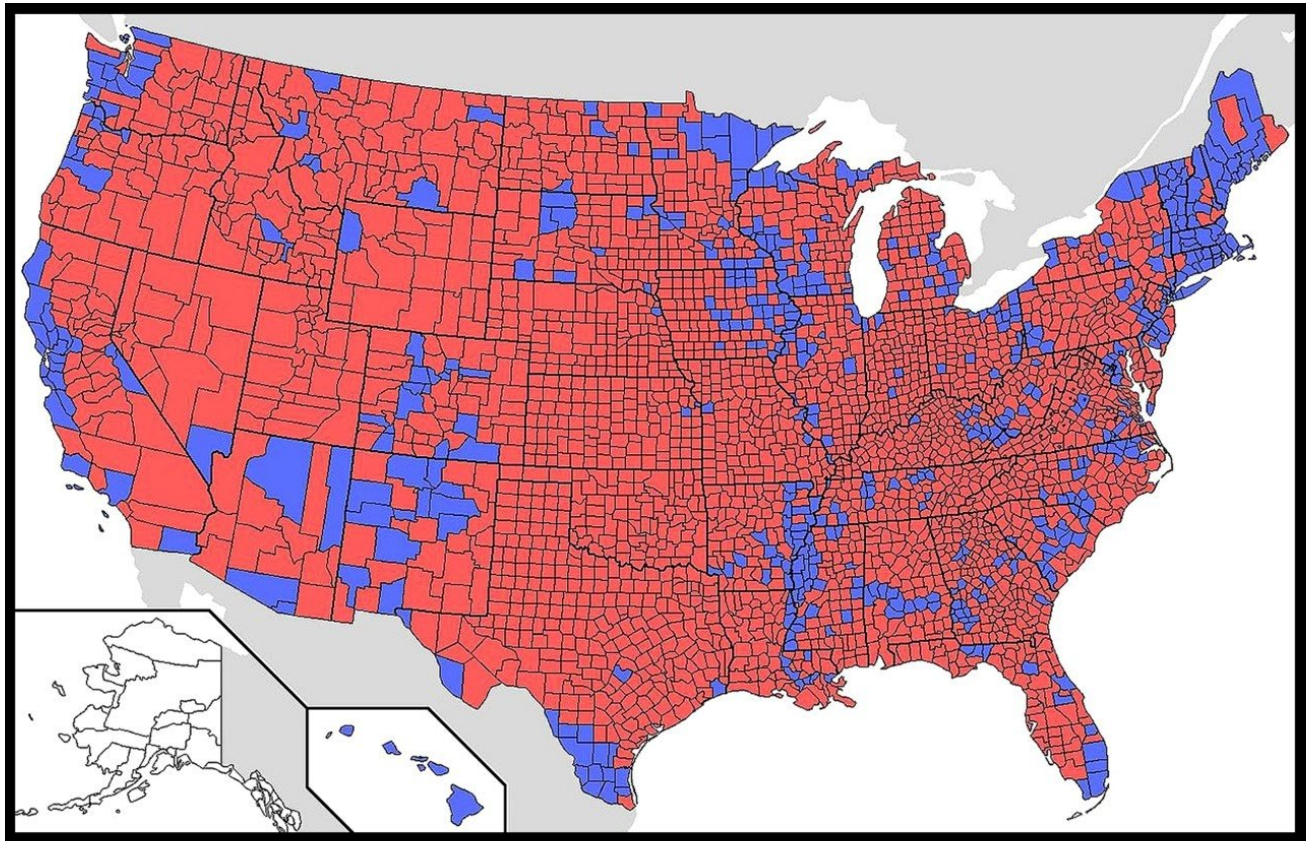
When you see the figure on the right, what do you interpret the colors as?

Answer: red is high unemployment



Last aside about colors: intuition

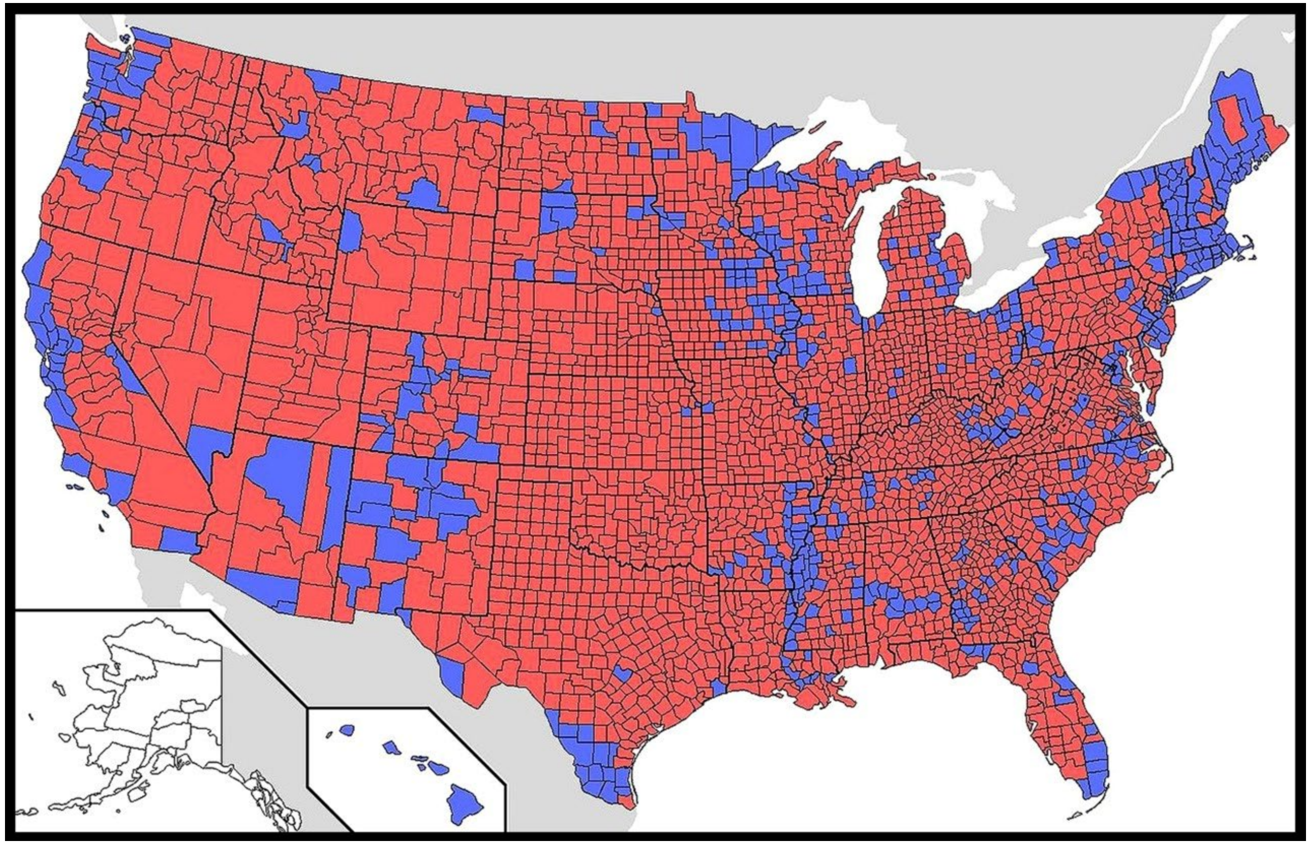
What about this one?



Last aside about colors: intuition

What about this one?

In the US, red and blue carry political connotations

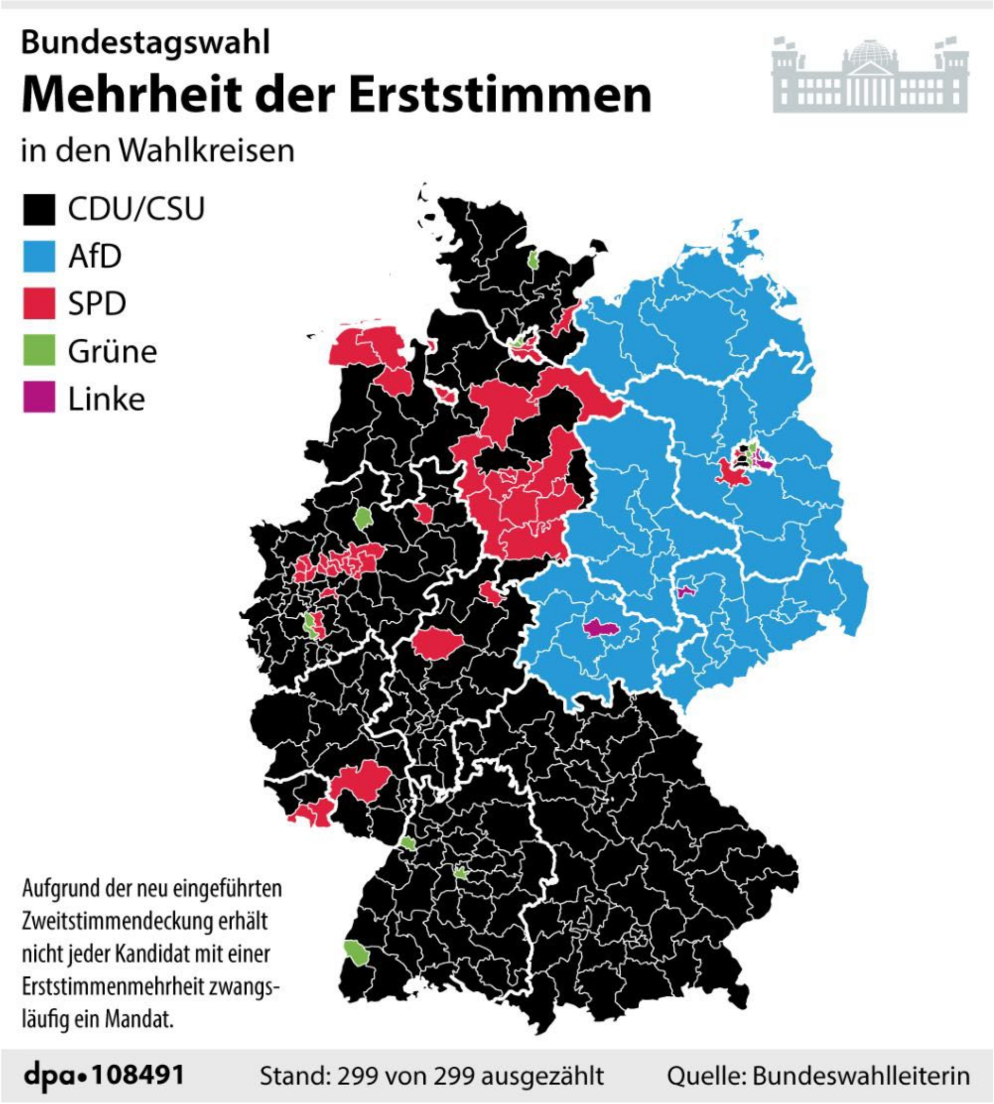


Last aside about colors: intuition

What about this one?

In the US, red and blue carry political connotations.

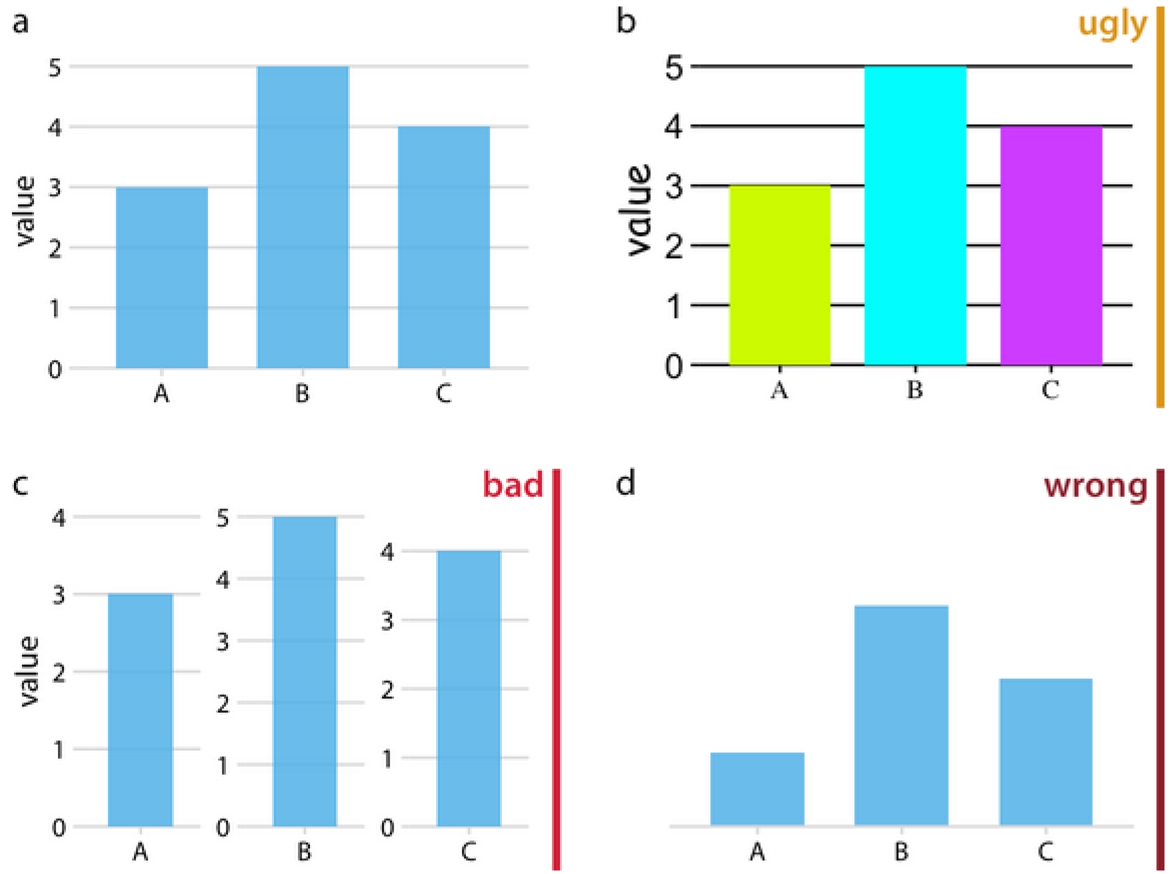
In another country, colors can map to politics very differently



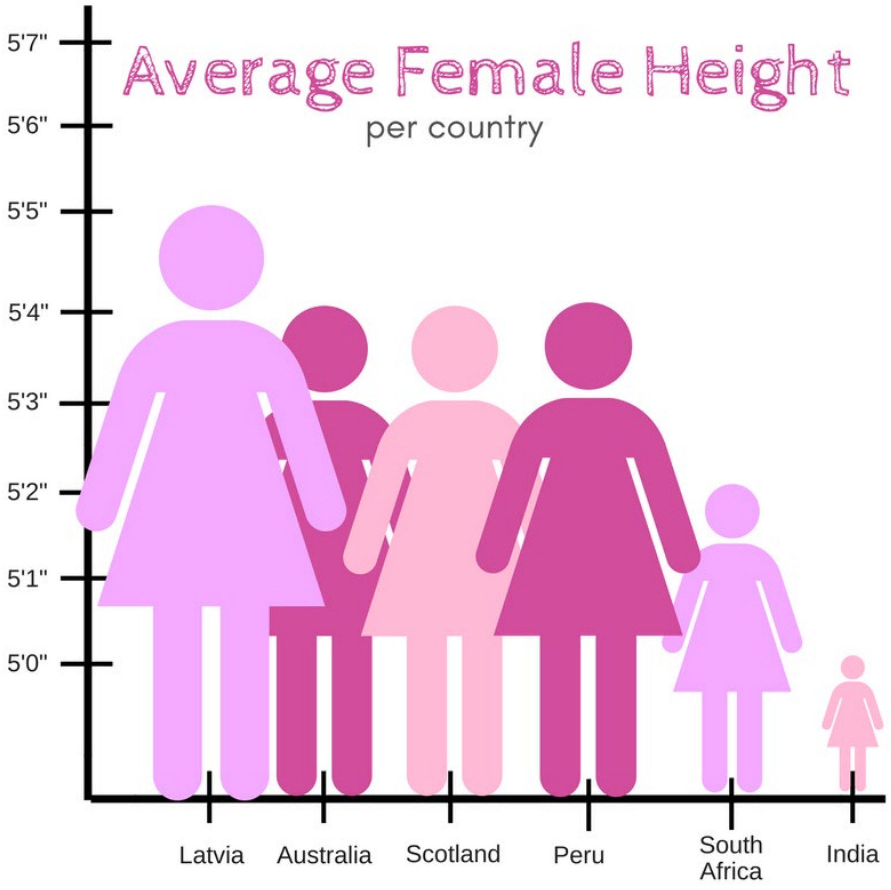
Wilke's "Cardinal Sins"

From the book *Fundamentals of Data Visualization:*

- ***Ugly***: *bad colors or aesthetically displeasing elements.*
- ***Bad:*** *not incorrect, but actively misleading.*
- ***Wrong:*** *contains incorrect information.*



Categorizing Bad Graphs: Good, bad, ugly, or wrong?



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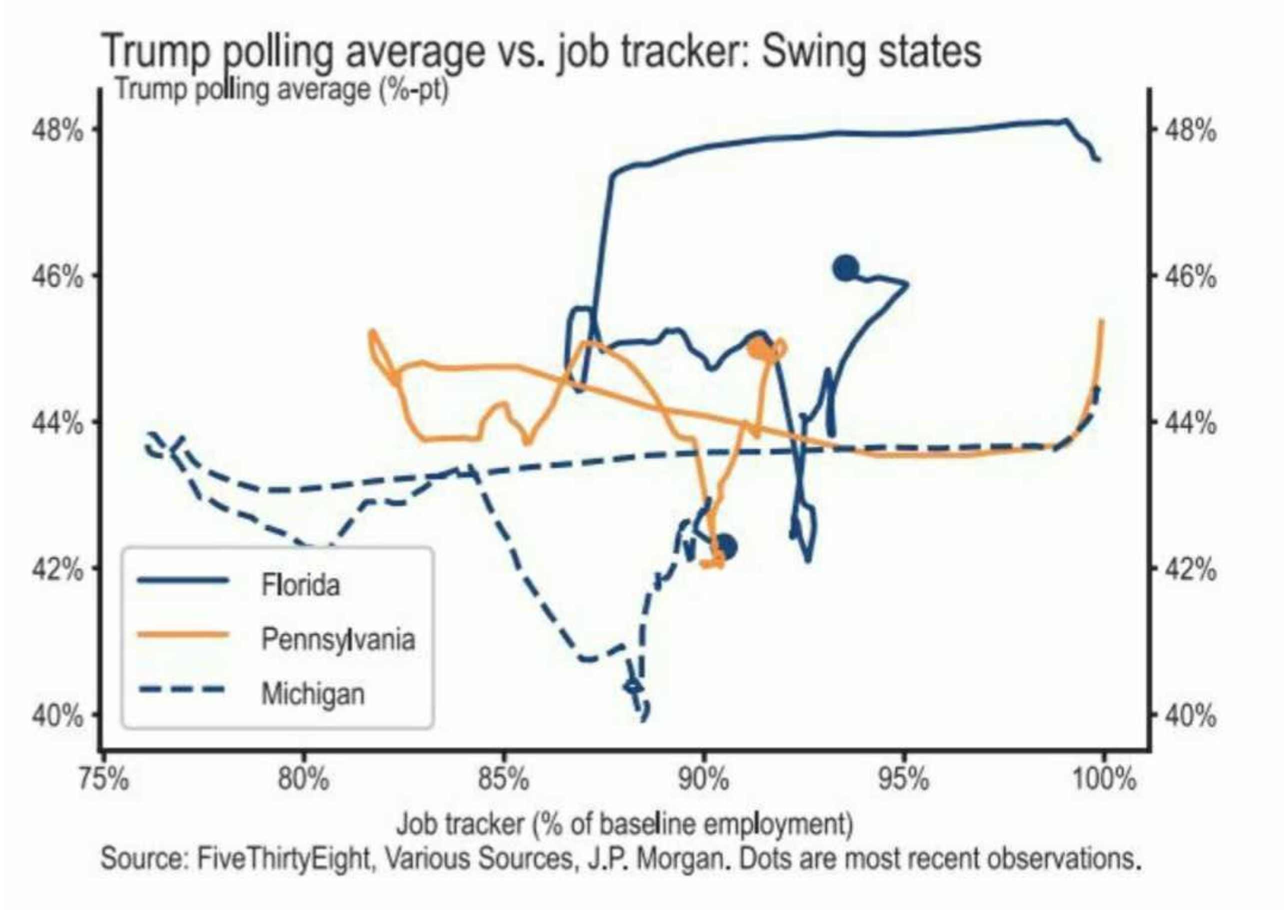
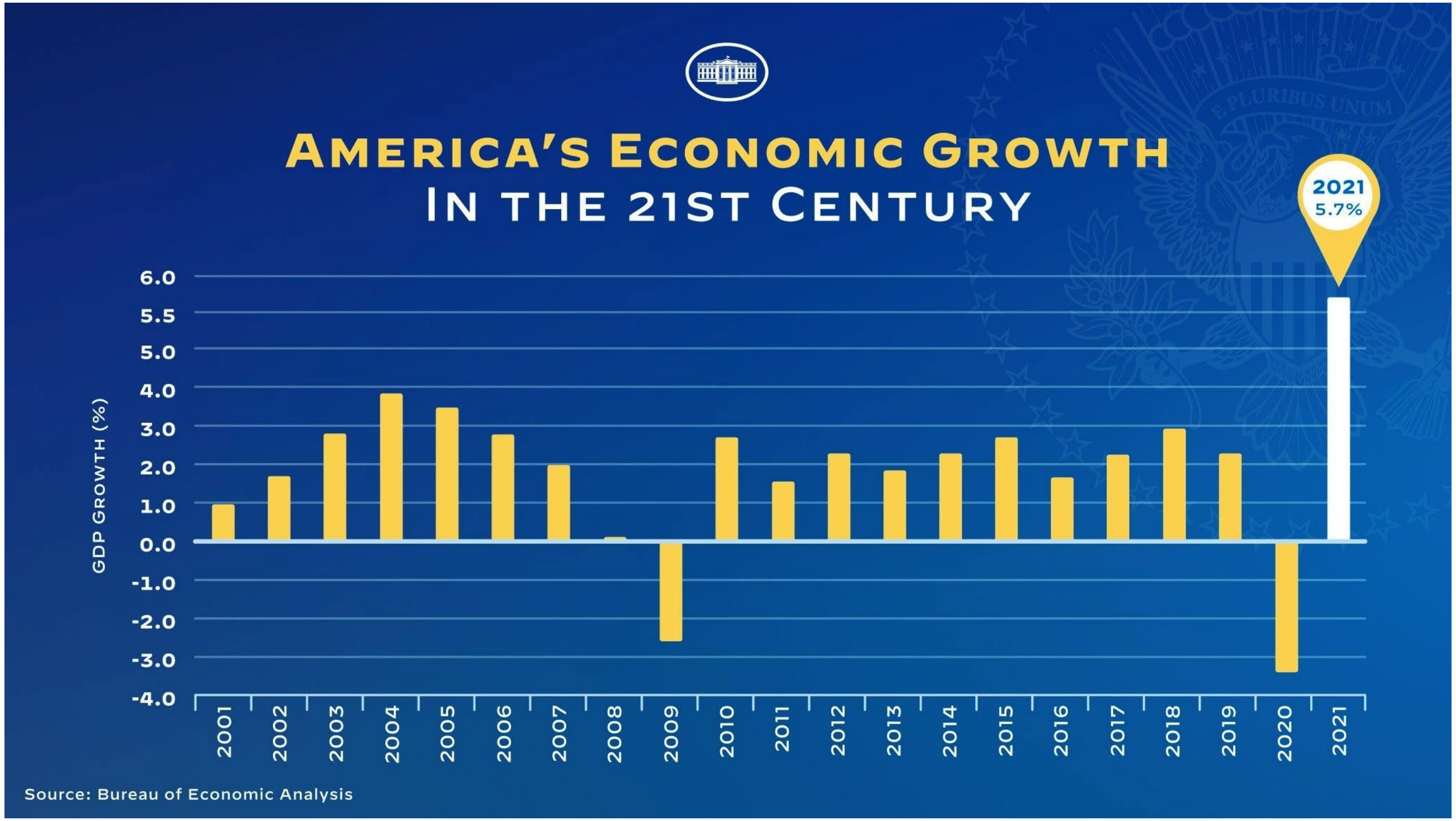
Good, bad, ugly, or wrong?



Categorizing Bad Graphs: Good, bad, ugly, or wrong?



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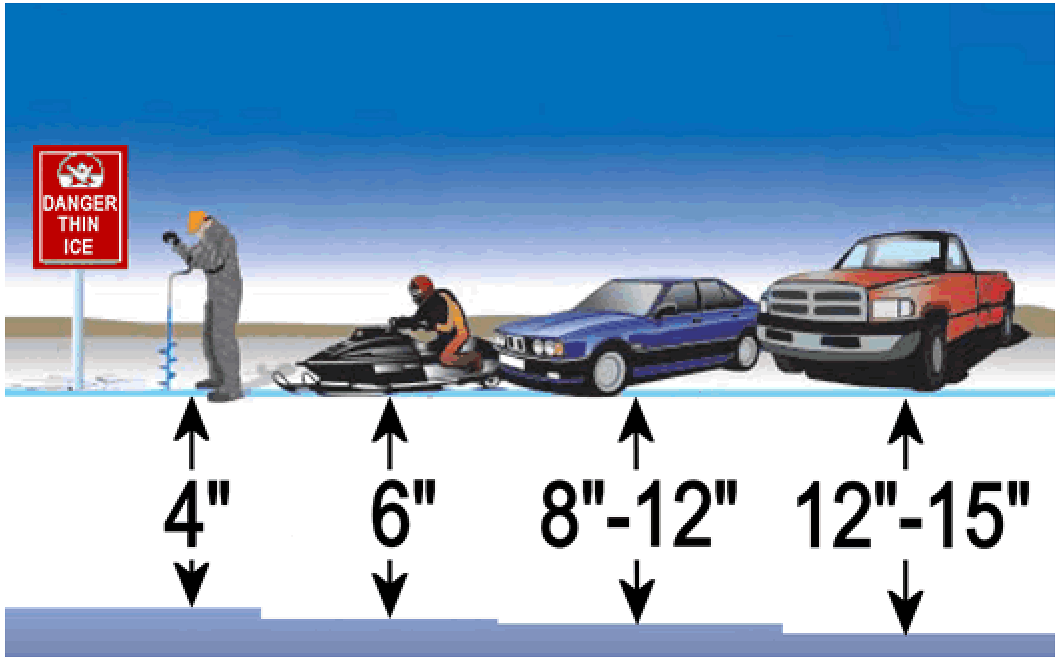
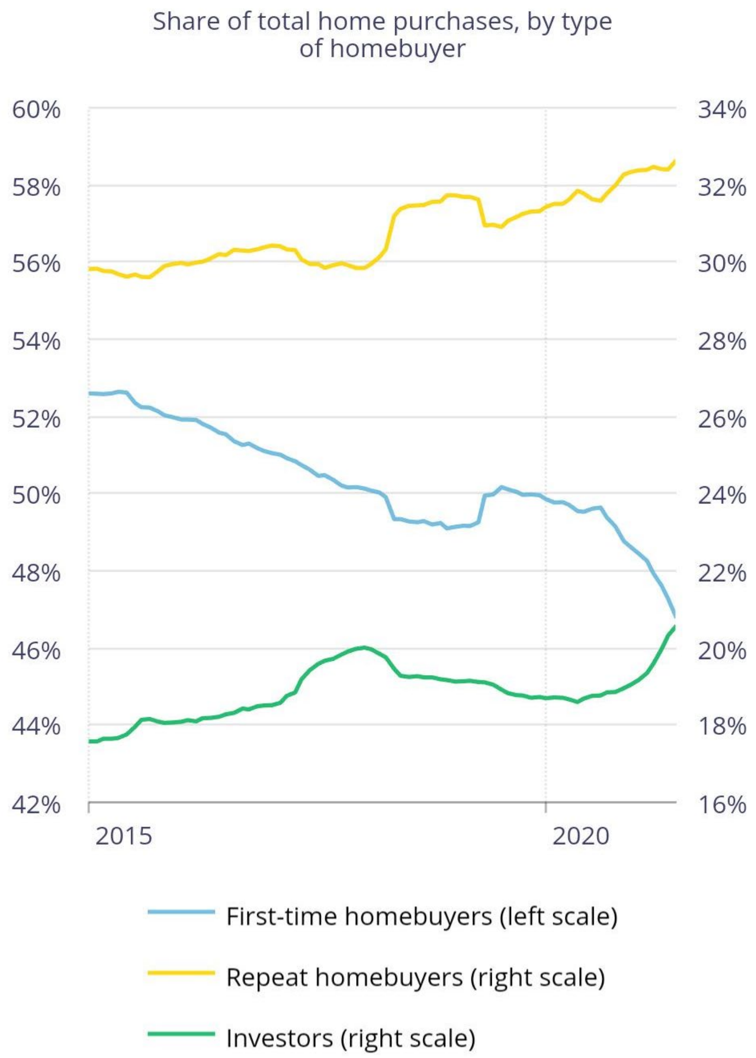


Categorizing Bad Graphs: Good, bad, ugly, or wrong?



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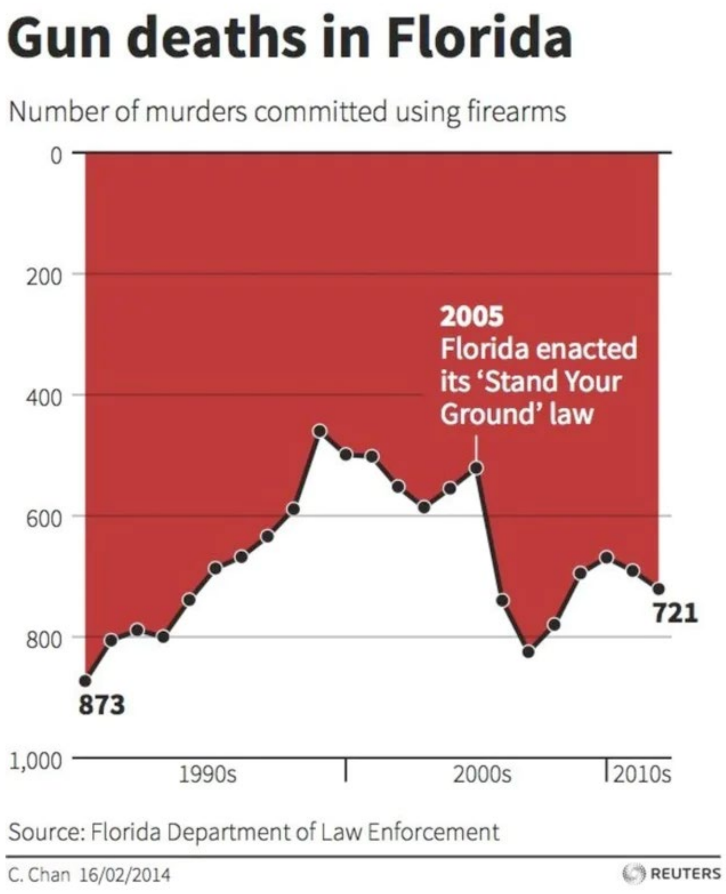

Categorizing Bad Graphs:
Good, bad, ugly, or wrong?